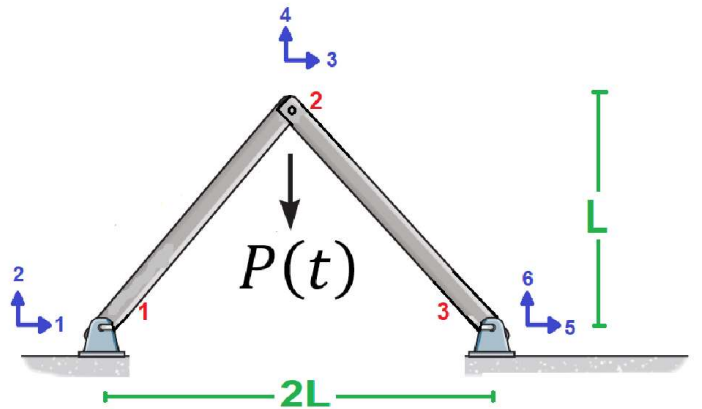


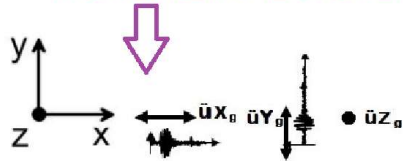
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

**COMPREHENSIVE NONLINEAR
SEISMIC ASSESSMENT OF TWO
ELEMENTS TRUSS: AN OPENSEES
FRAMEWORK FOR STATIC
PUSHOVER, CYCLIC DEGRADATION,
STATIC TIME-HISTORY AND
DYNAMIC TIME-HISTORY ANALYSIS**

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)



INCREMENTAL DISPLACEMENT



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

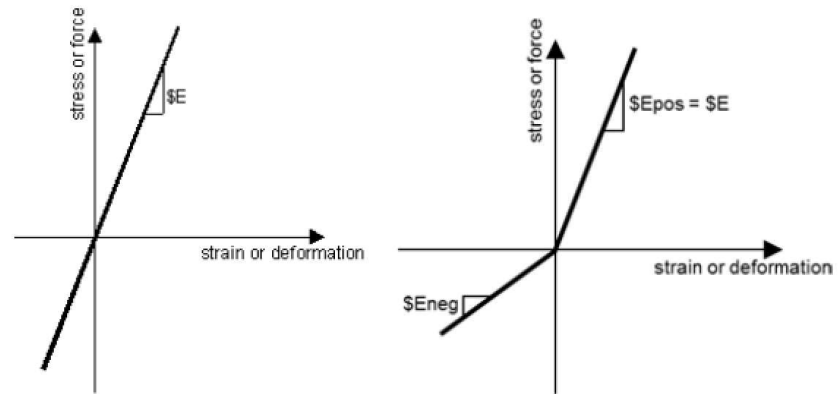
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

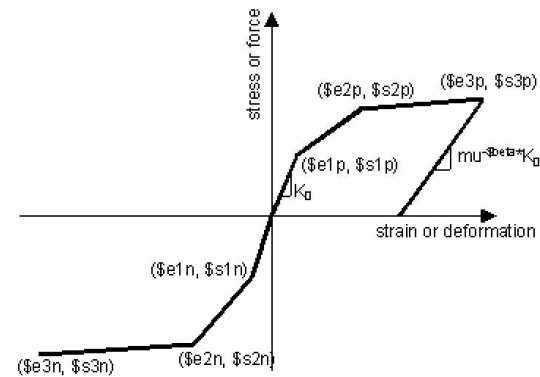
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



INELASTIC MATERIAL

Spyder (Python 3.12)

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...\26_TRUSS_TWO_ELEMENTS\P(t)_TRUSS_TWO_ELEMENTS.py

P(t)_TRUSS_TWO_ELEMENTS.py

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF TWO ELEMENTS TUSS : AN OPENSEES
4 # FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-H
5 #
6 # W(x,t) = W0 sin(wt)
7 # W(x,t) = W0 exp(-0.05wt) sin(wt)
8 #
9 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND E
10 # OF ENERGY DISSIPATION CAPACITY INDEX
11 #
12 # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
13 # EMAIL: salar.d.ghashghaei@gmail.com
14 #####
15 """
16 Nonlinear Seismic Performance Assessment of Two Elements Truss:
17 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and
18
19 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of Two Elem
20 for performance-based earthquake engineering. The 2D model incorporates both material non
21 (elastic-perfectly plastic or hysteretic steel with strain hardening)
22 and geometric nonlinearity to capture P-delta effects
23 and large displacements-essential for collapse assessment.
24
25 Eight analysis protocols are implemented:
26 (1) [PERIOD] : Structural Period
27 (2) [STATIC] : Gravity load analysis establishing dead load state
28 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
29 and plastic mechanism identification
30 (4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
31 pinching behavior, and energy dissipation degradation
32 (5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External time-dependent
33 (6) [DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Dynamic Analysis of External time-dependen
34 (7) [FREE-VIBRATION] : Free-vibration with initial conditions extracting damping ratios
```

Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 19, Col 101 UTF-8 CRLF RW Mem 75%

STATIC ANALYSIS

Spyder (Python 3.12)

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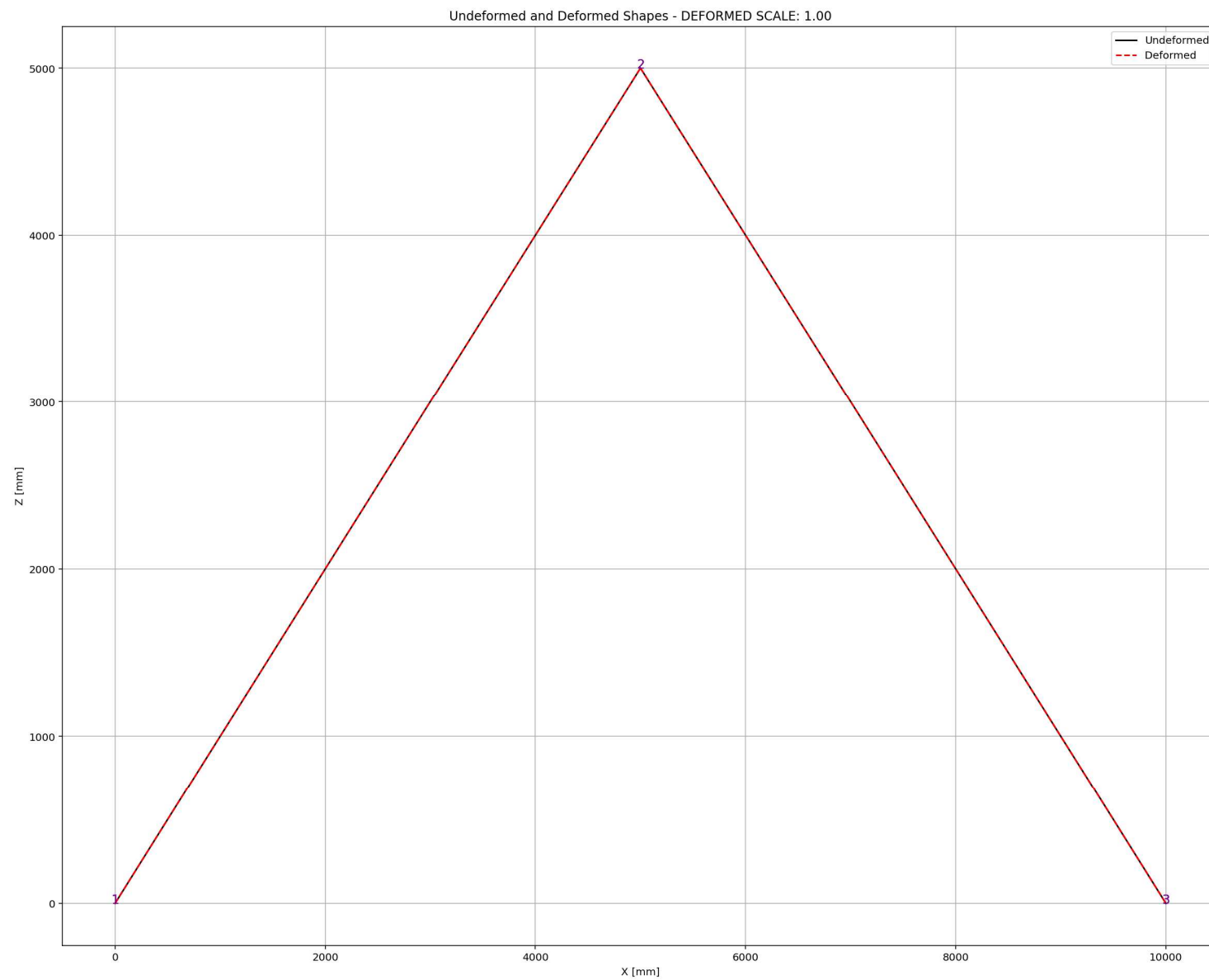
P(t)_TRUSS_TWO_ELEMENTS.py

```
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Underformed and Deformed Shapes - DEFORMED SCALE: 1.00

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 19, Col 101 UTF-8 CRLF RW Mem 75%



PUSHOVER ANALYSIS

Spyder (Python 3.12)

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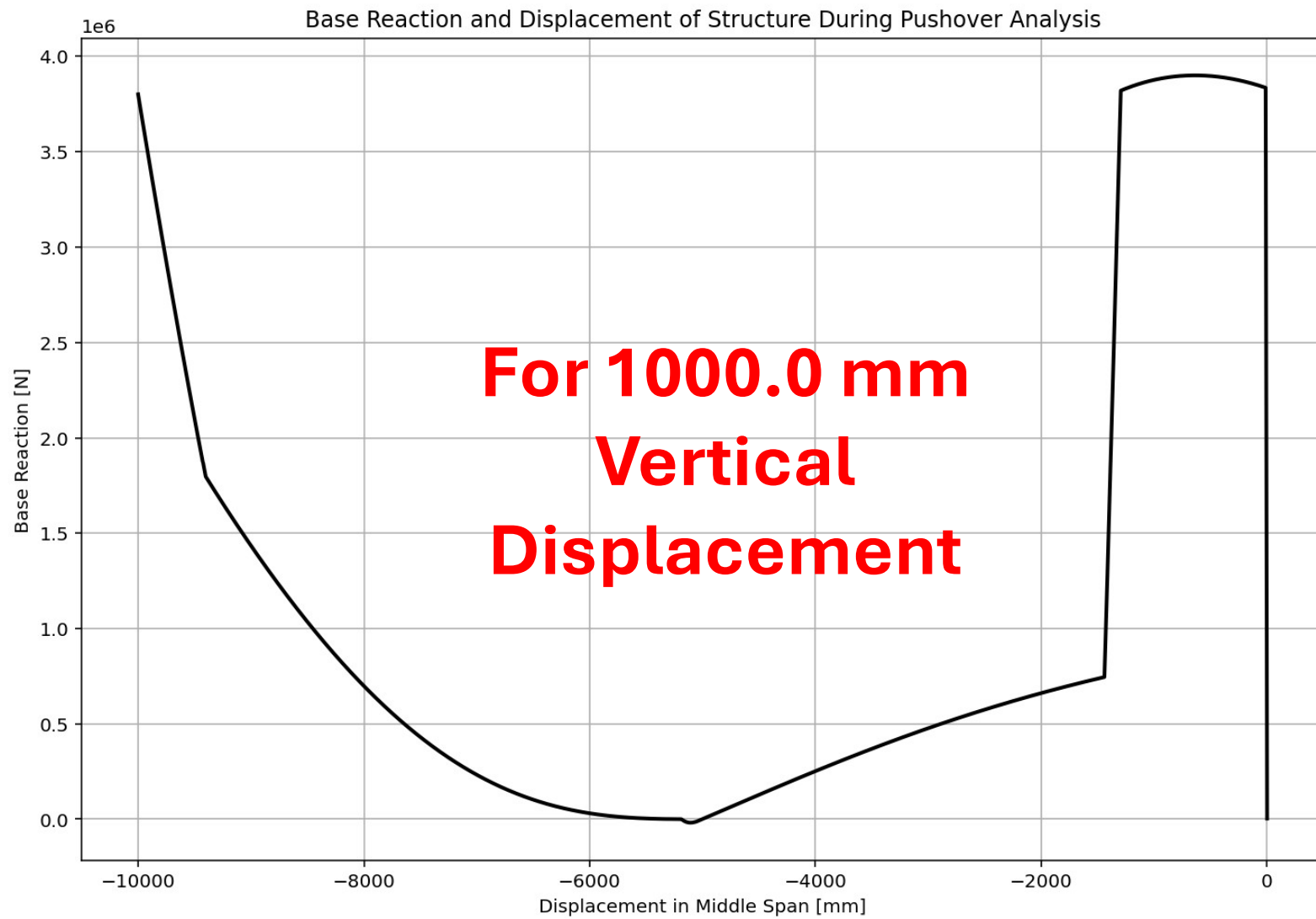
P(t)_TRUSS_TWO_ELEMENTS.py X

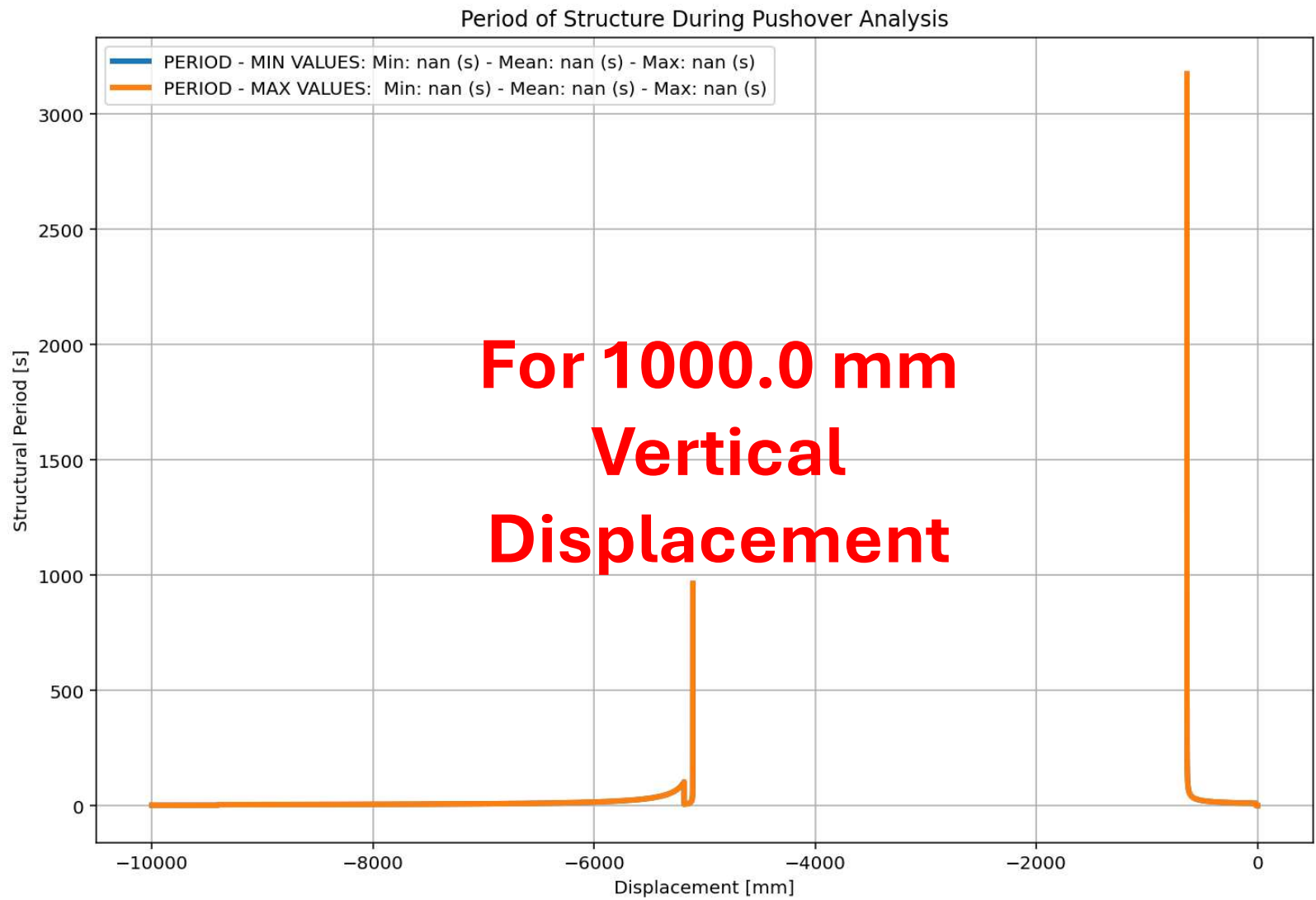
```
278
279 if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
280     IDctrlDOF = 2 # 1: Horizontal Dispalcement - 2: Vertical Dispalcement
281     DINCR = -0.01 # [mm] Incremental Vertical Displacement
282     DMAX = -10000.0 # [mm] Max. Vertical Displacement
283     ops.integrator('DisplacementControl', center_node, IDctrlDOF, DINCR)
284     ops.analysis('Static')
285     Nsteps = int(np.abs(DMAX/ DINCR))
286     STEP = 0.0
287     for step in range(Nsteps):
288         OK = ops.analyze(1)
289         S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
290         ops.reactions()
291         reaction.append(ops.nodeReaction(1, 2)+ops.nodeReaction(3, 2)) # SHEAR BASE R
292         disp_mid.append(ops.nodeDisp(center_node, 2)) # DISPLACEMENT
293         dispX.append(ops.nodeDisp(center_node, 1)) # DISPLACEMENT
294         dispY.append(ops.nodeDisp(center_node, 2)) # DISPLACEMENT
295         # Store elements force, strain and stress
296         for ele_id in ele_axialforce.keys():
297             ele_axialforce[ele_id].append(ops.eleResponse(ele_id, 'axialForce')) #
298             ele_stress[ele_id].append(ops.eleResponse(ele_id, 'material', 'stress')) #
299             ele_strain[ele_id].append(ops.eleResponse(ele_id, 'material', 'strain')) #
300             if ELE_TYPE == 'INELASTIC':
301                 ele_di[ele_id].append(100*(abs(ele_strain[ele_id]) - ey) - (esu - ey))
302             if ELE_TYPE == 'ELASTIC':
303                 ele_di[ele_id].append(0.0)
304         # Store displacements
305         for node_id in node_displacementsX.keys():
306             node_displacementsX[node_id].append(ops.nodeDisp(node_id, 1))
307             node_displacementsY[node_id].append(ops.nodeDisp(node_id, 2))
308         # IN EACH STEP, STRUCTURE PERIOD GOING TO BE CALCULATED
309         #PERIODmin, PERIODmax = S06.RAYLEIGH_DAMPING(1, 0.5*DR, DR, 0, 1)
310         PERIODmin, PERIODmax = S05.EIGENVALUE_ANALYSIS(1, PLOT=True)
311         PERIOD_MIN.append(PERIODmin)
```

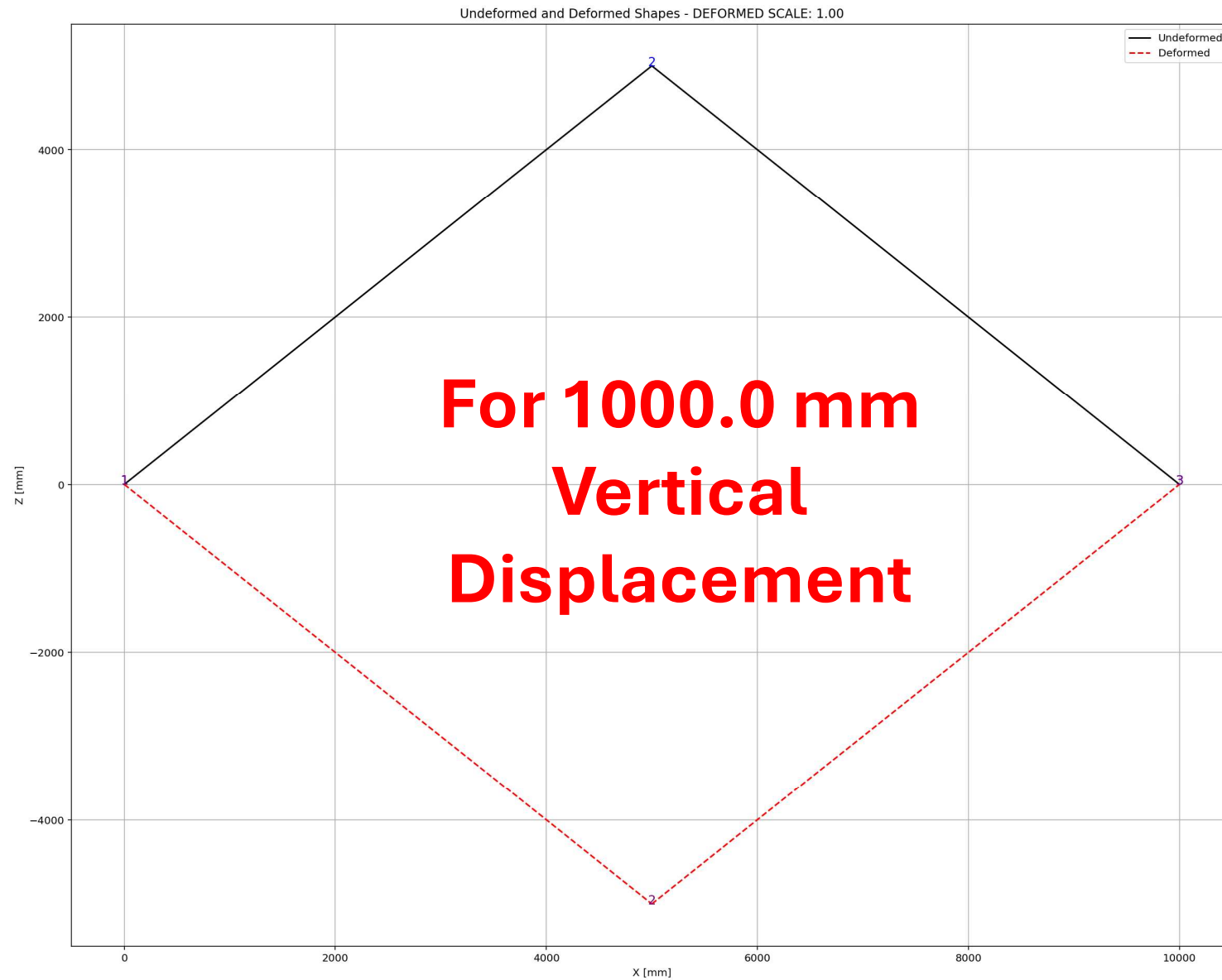
Base Reaction and Displacement of Structure During Pushover Analysis

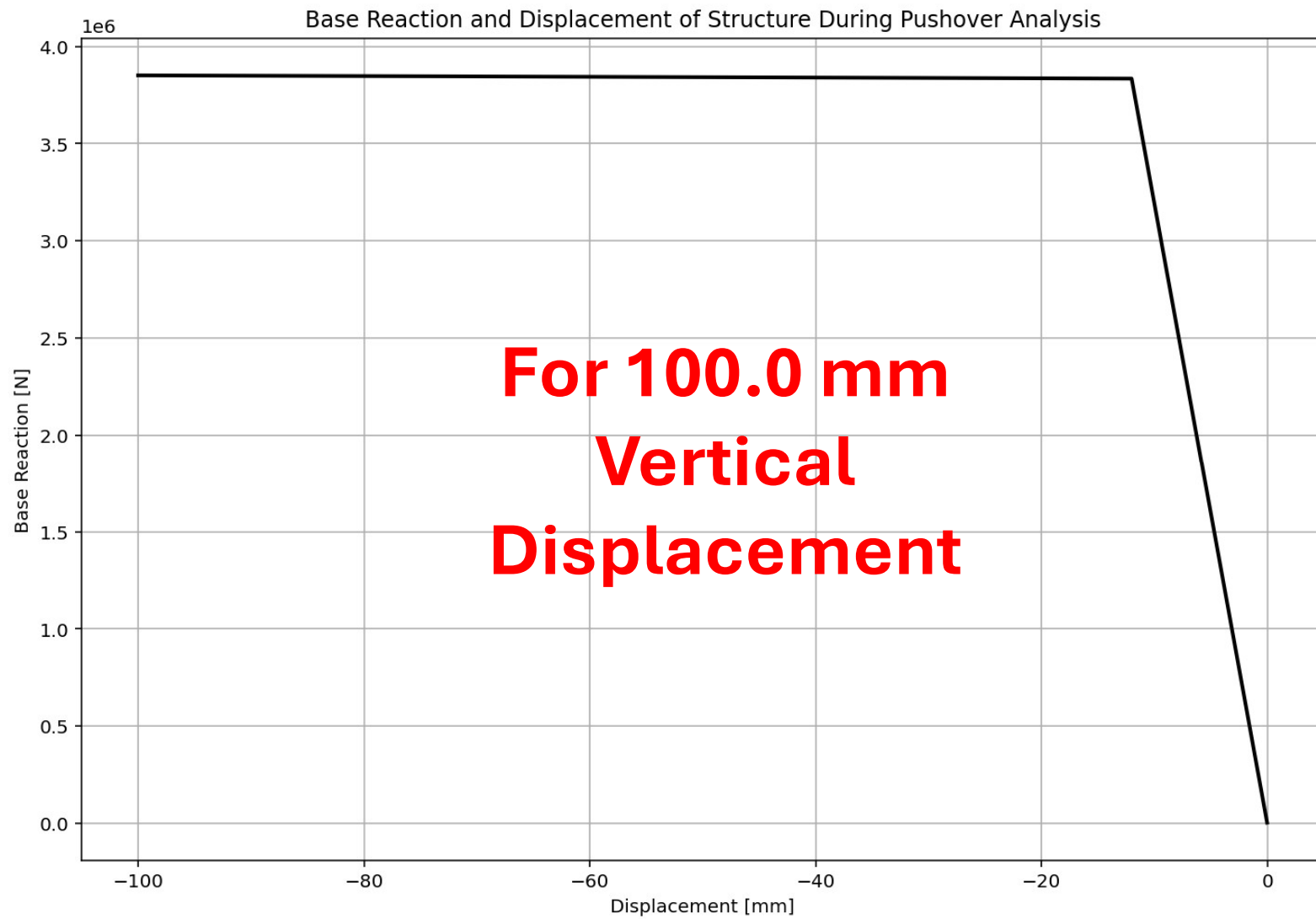
IPython Console Files Help Variable Explorer Debugger Plots History

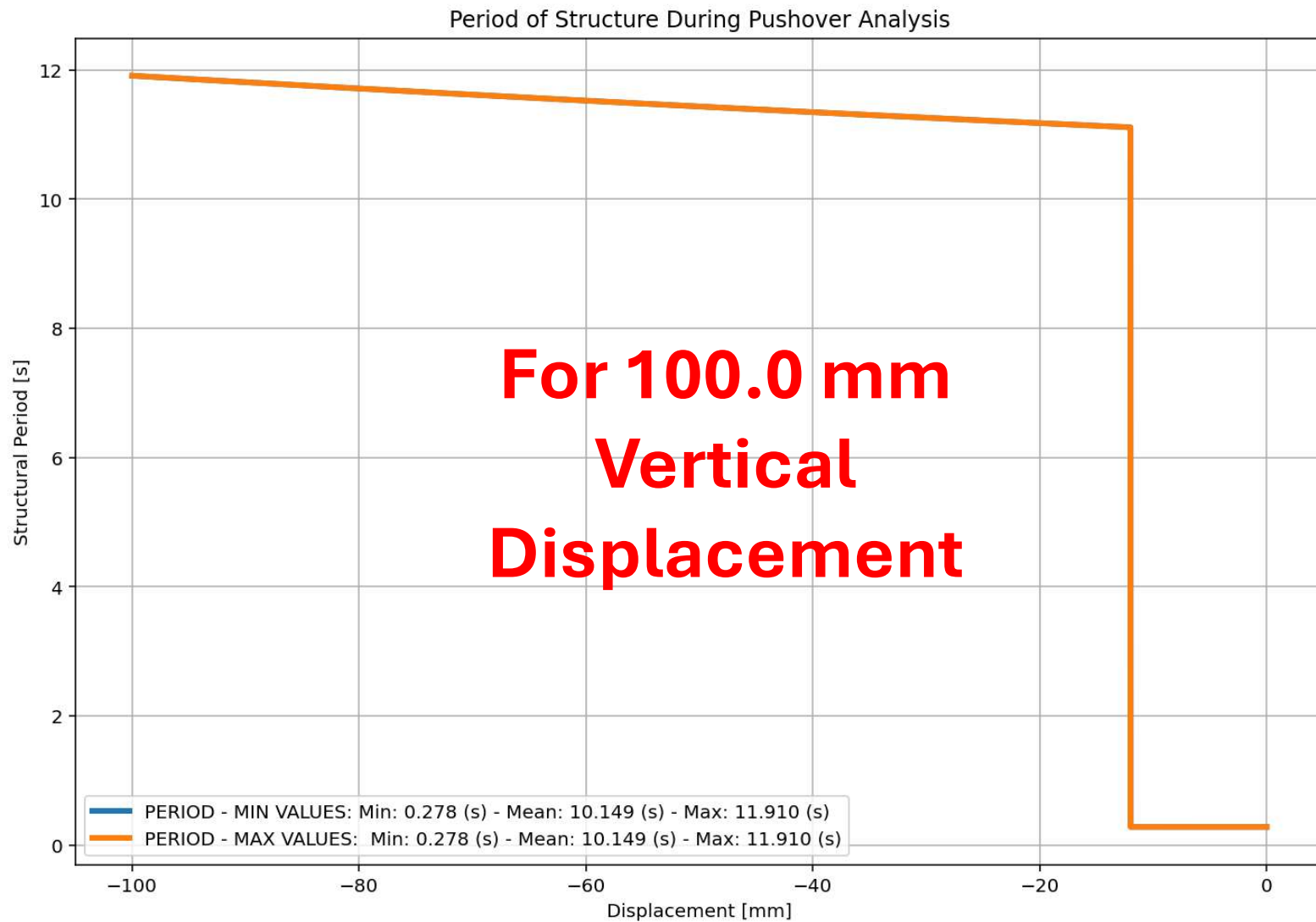
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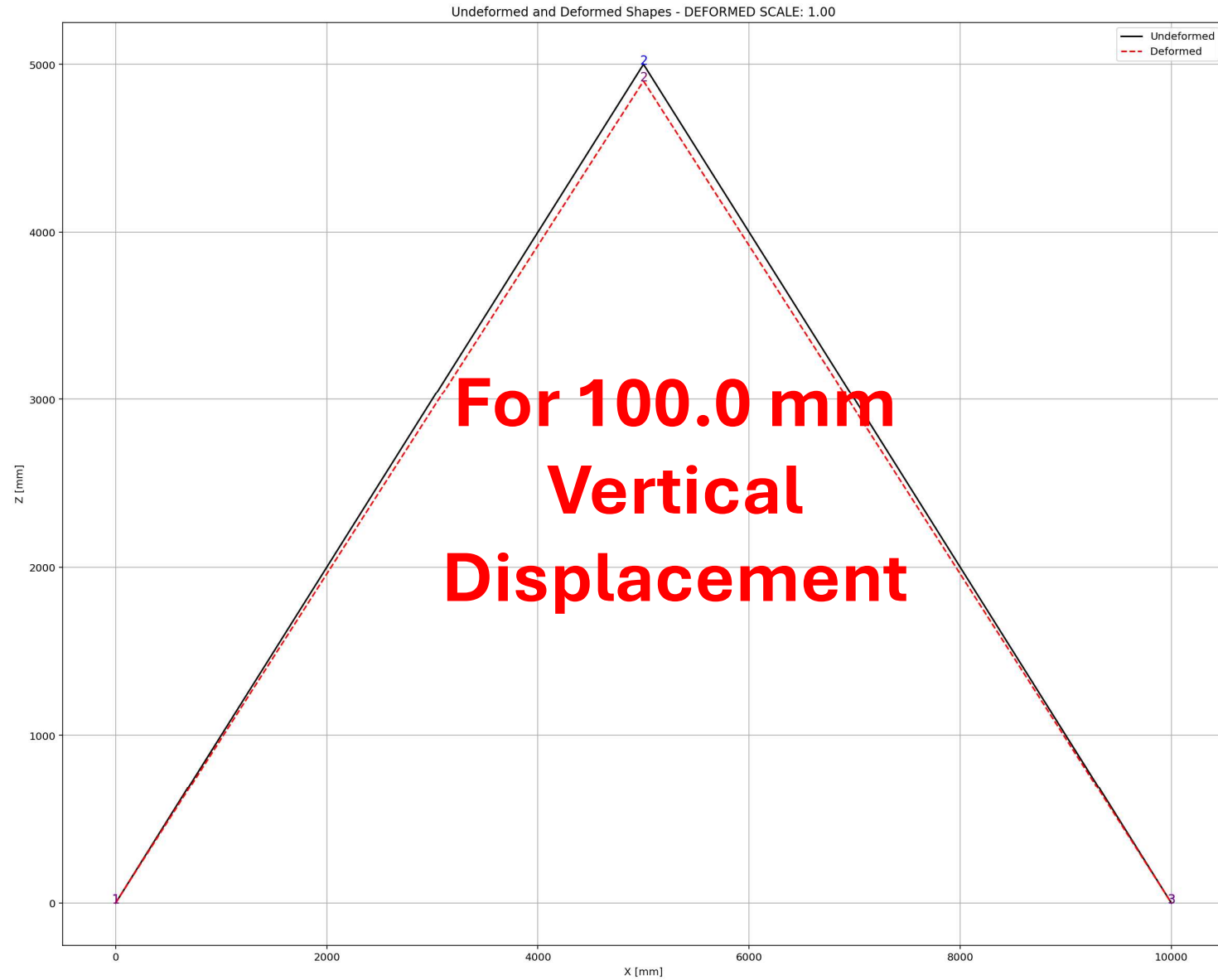












CYCLIC DISPLACEMENT ANALYSIS

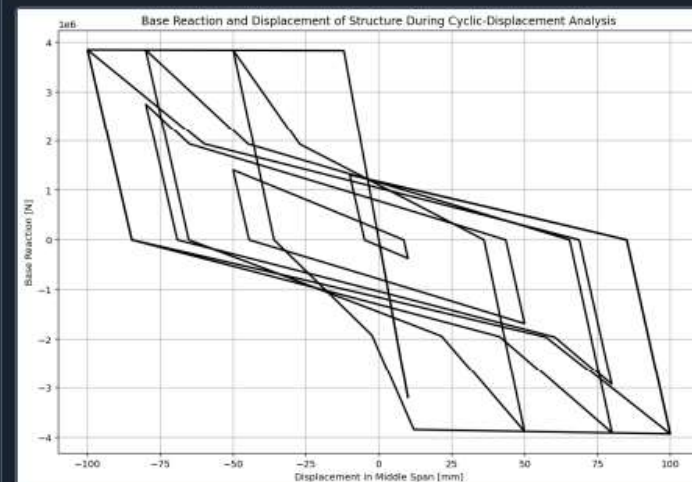
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...\26_TRUSS_TWO_ELEMENTS\P(t)_TRUSS_TWO_ELEMENTS.py

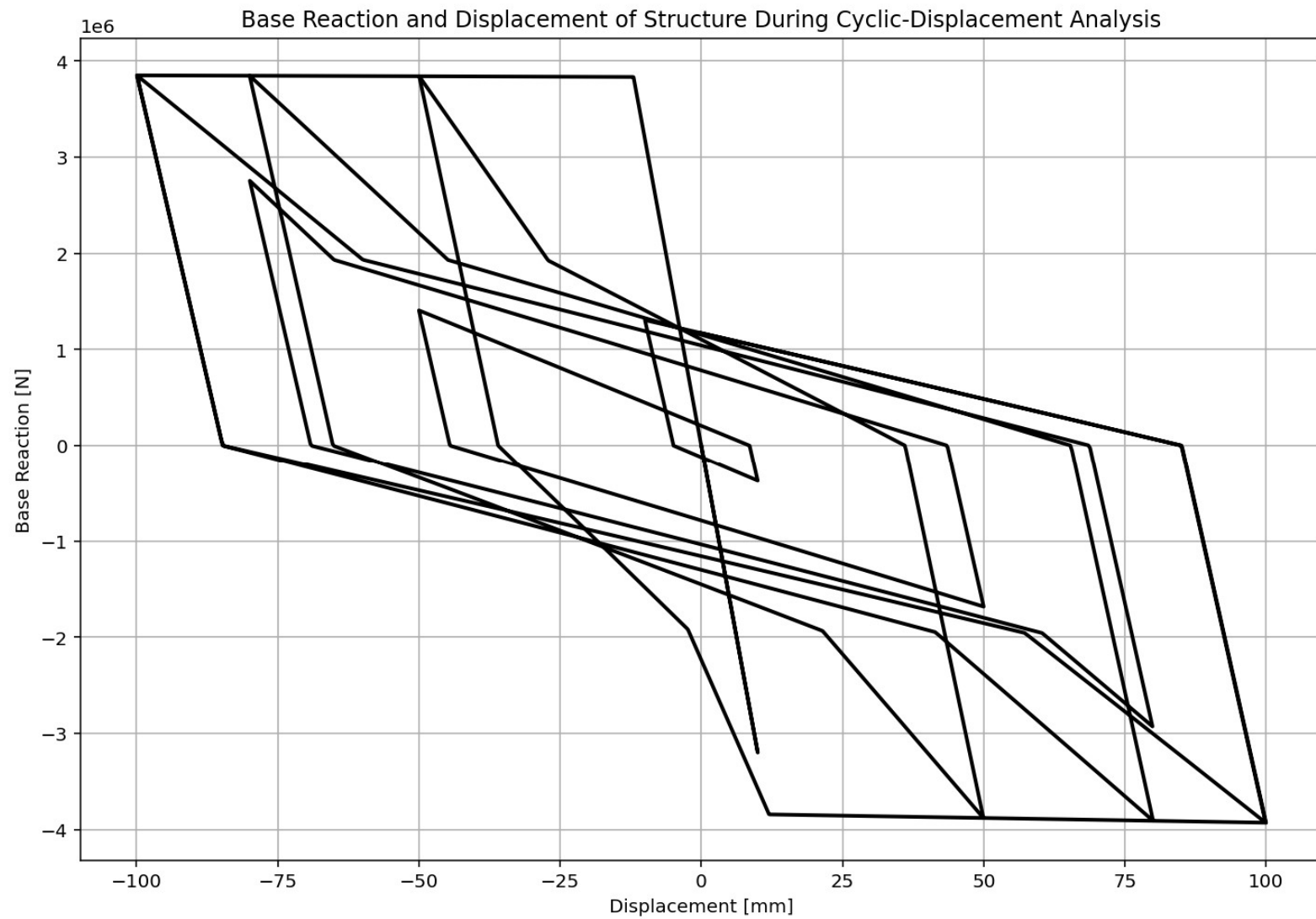
P(t)_TRUSS_TWO_ELEMENTS.py X

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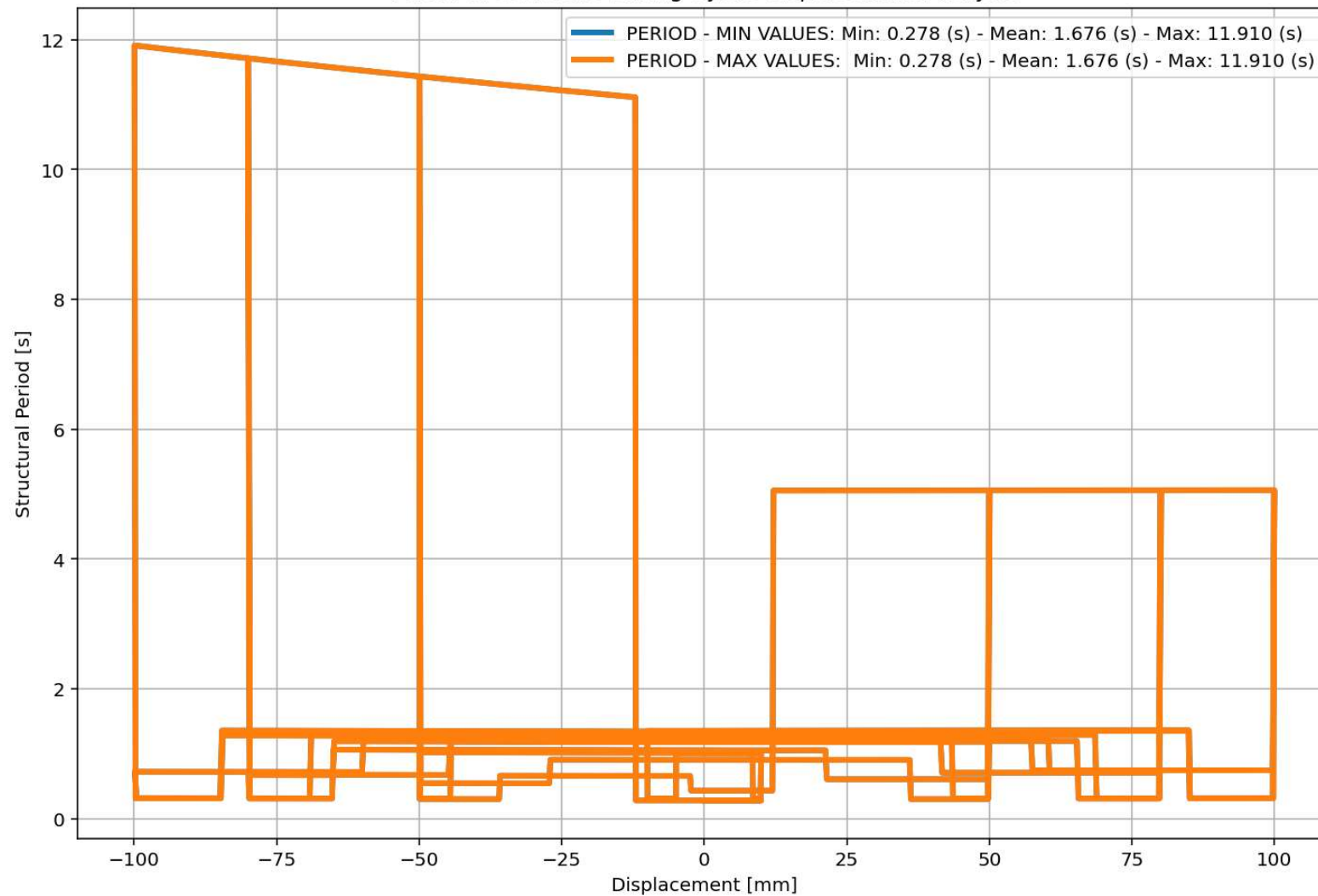
327 if ANAL_TYPE == 'CYCLIC_DISPLACEMENT': # STATIC TIME-HISTORY ANALYSIS
328     IDctrlDOF = 2 # 1: Horizontal Displacement - 2: Vertical Displacement
329     DMAX = -100.0 # [mm] Max. Vertical Displacement
330     n_points = 10000
331     # 4. CYCLIC DISPLACEMENT PROTOCOL
332     # Key strain points (same logic as your protocol)
333     key_disp = np.array([
334         0.0,
335         0.1*DMAX, -0.1*DMAX,
336         0.5*DMAX, -0.5*DMAX,
337         0.8*DMAX, -0.8*DMAX,
338         DMAX, -DMAX,
339         0.1*DMAX, -0.1*DMAX,
340         0.5*DMAX, -0.5*DMAX,
341         0.8*DMAX, -0.8*DMAX,
342         DMAX, -DMAX,
343         0.0
344     ])
345
346     # Generate 1000-point displacement protocol
347     disp_protocol = np.interp(
348         np.linspace(0, len(key_disp) - 1, n_points),
349         np.arange(len(key_disp)),
350         key_disp
351     )
352
353     ops.analysis('Static')
354     STEP = 0.0
355     for target_disp in disp_protocol:
356         current_disp = ops.nodeDisp(center_node, 2) # DISPALCEMENT APPLIED IN MIDDLE N
357         dU = target_disp - current_disp
358         ops.integrator('DisplacementControl', center_node, IDctrlDOF, dU)
359         OK = ops.analyze(1)
360         S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)

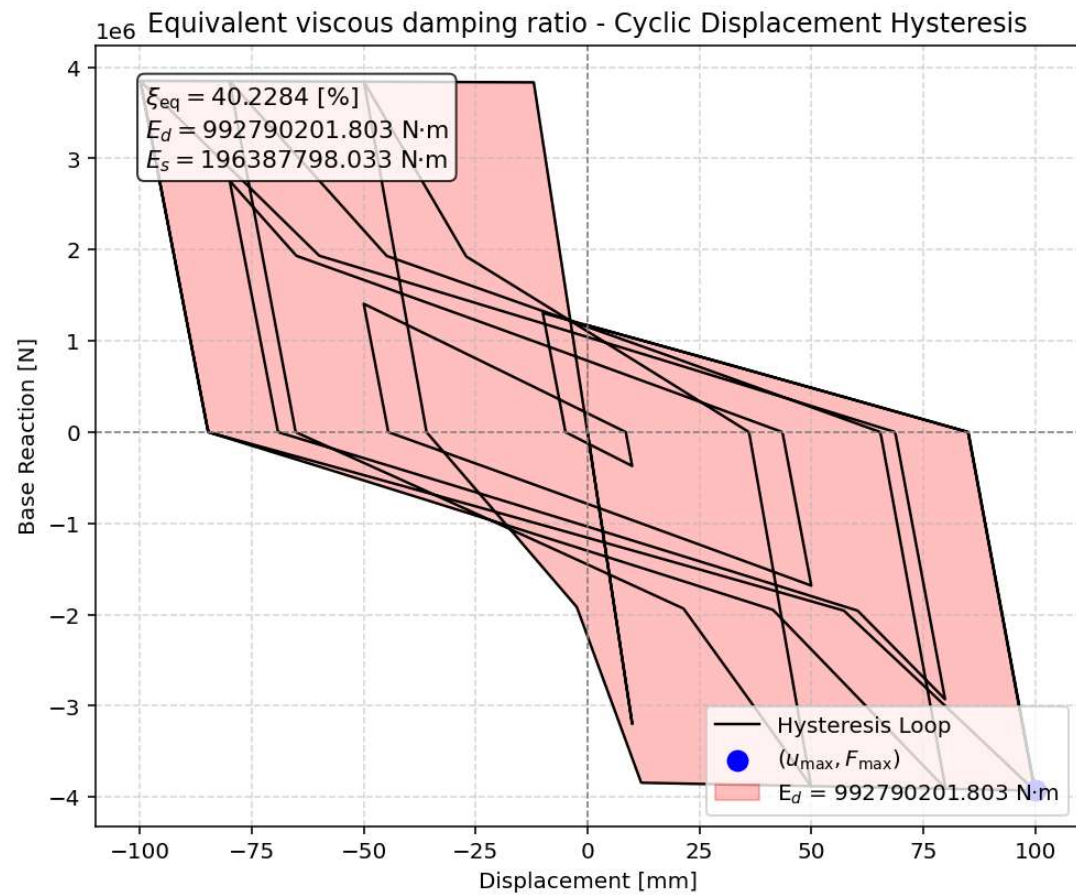
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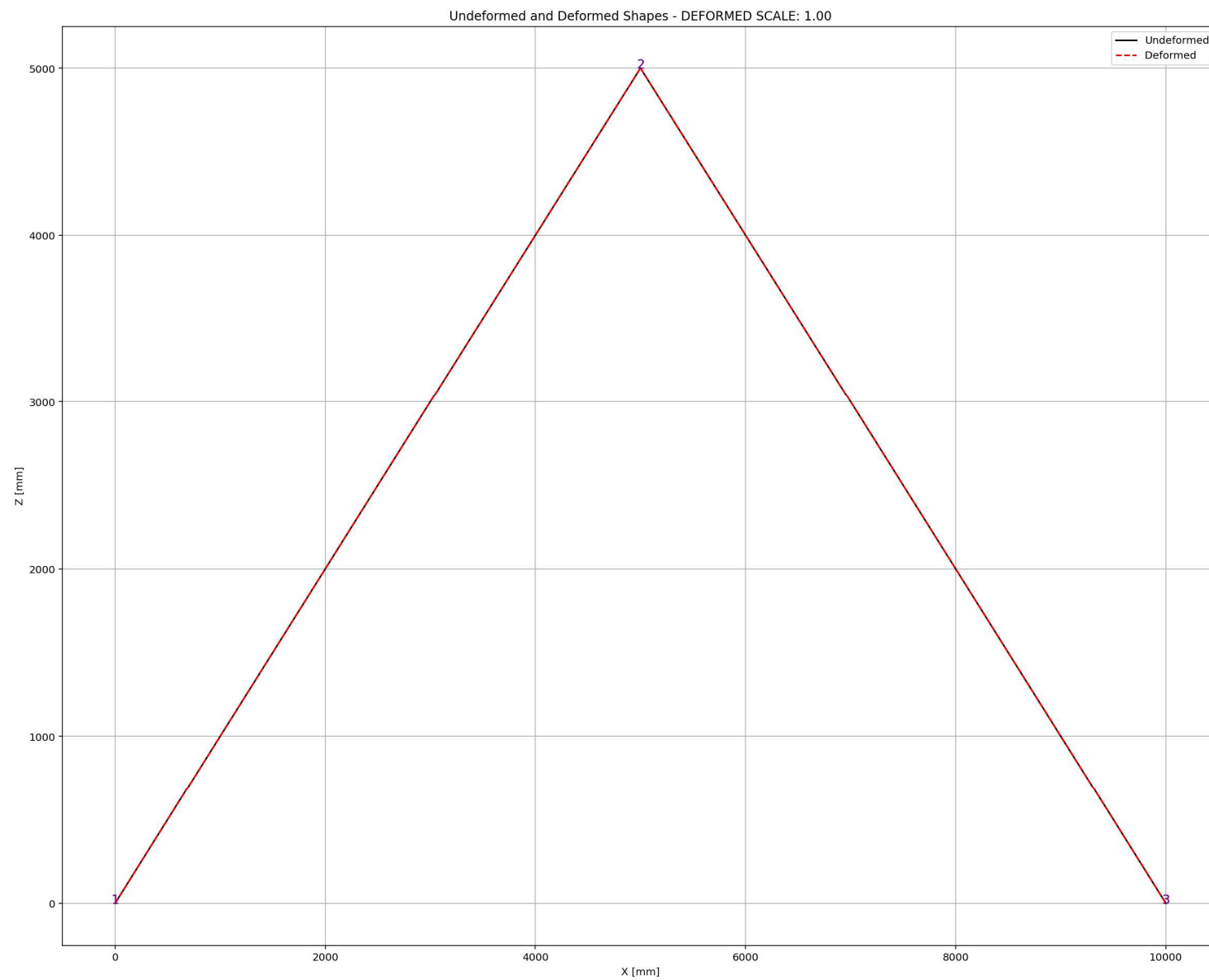




Period of Structure During Cyclic Displacement Analysis







STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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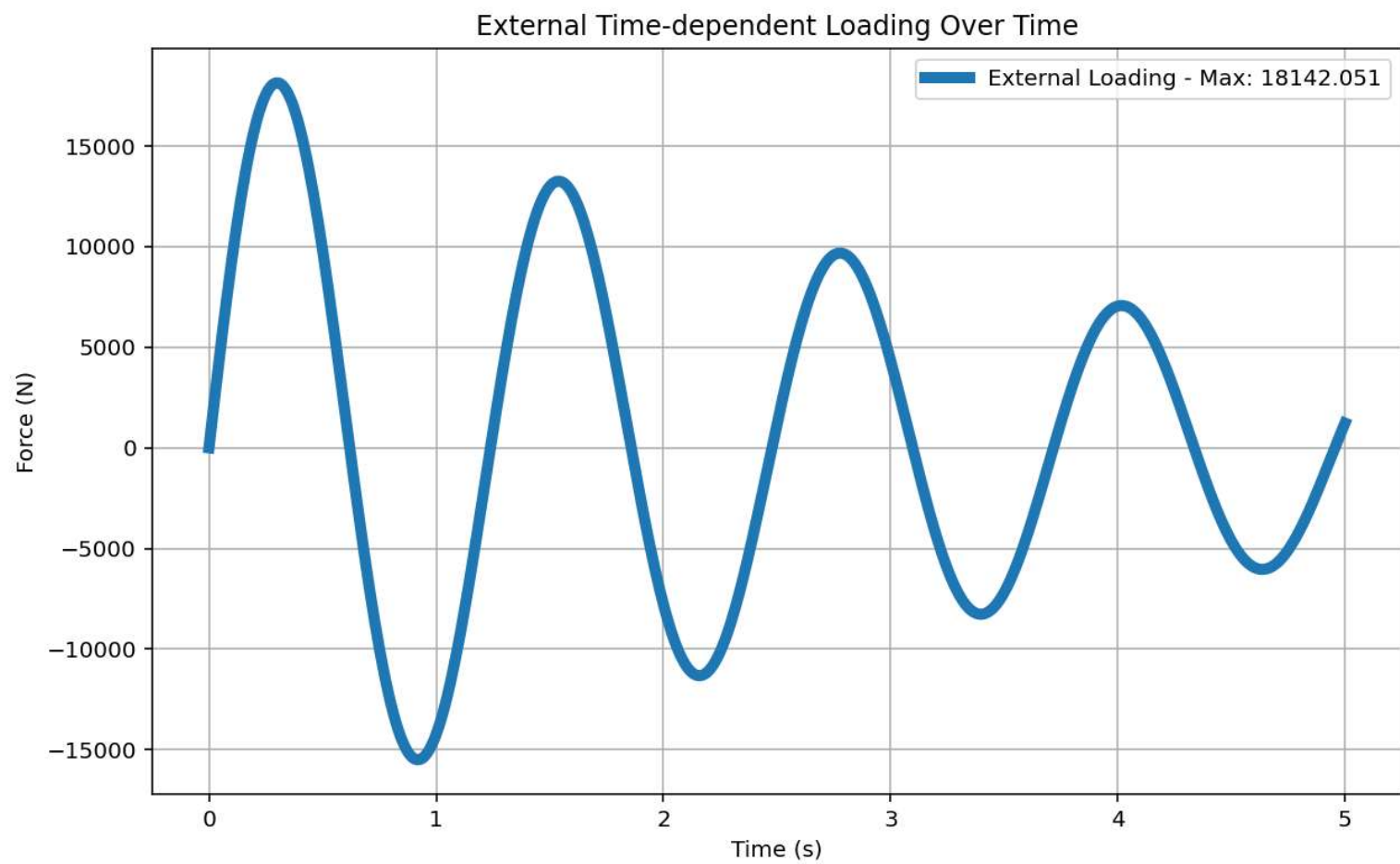
P(t)_TRUSS_TWO_ELEMENTS.py

```
398 if ANAL_TYPE == 'STATIC EXTERNAL TIME-DEPENDENT LOADING': # STATIC TIME-HISTORY ANALYSIS
399     %% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
400     # IN HERE ANALYSIS TIME AND DURATION ARE LOAD STEPS
401     duration = 20.0 # [s] Analysis duration
402     dt = 0.01 # [s] Time step
403     DT = dt # [s] Time step
404     DT_time = 5.0 # [s] Total external Load Analysis Durations [*****]
405     force_amplitude = 19600.0 # [N] Amplitude Force
406     omega_DT = 5.0715 # [rad/s] Natural angular frequency
407     time_steps = int(duration/dt)
408
409     # Check function
410     def CHECK_FUN(DT_time ,duration):
411         if DT_time > duration:
412             print('\n\nAnalysis Duration Must be greater than External Load Duration\n')
413             exit()
414         return -1
415
416     CHECK_FUN(DT_time ,duration)
417     def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 sin
418         import numpy as np
419         import matplotlib.pyplot as plt
420         # External Load Durations
421         num_steps = int(DT_time / DT)
422         load_time = np.linspace(0, DT_time, num_steps)
423         target_frequency = 1.0 * omega_DT # Target excitation frequency
424         DT_load = force_amplitude * np.sin(target_frequency * load_time)
425         # Plot External Time-dependent Loading
426         plt.figure(figsize=(10, 6))
427         plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_load)}')
428         plt.title('External Time-dependent Loading Over Time')
429         plt.xlabel('Time (s)')
430         plt.ylabel('Force (N)')
431         plt.grid(True)
```

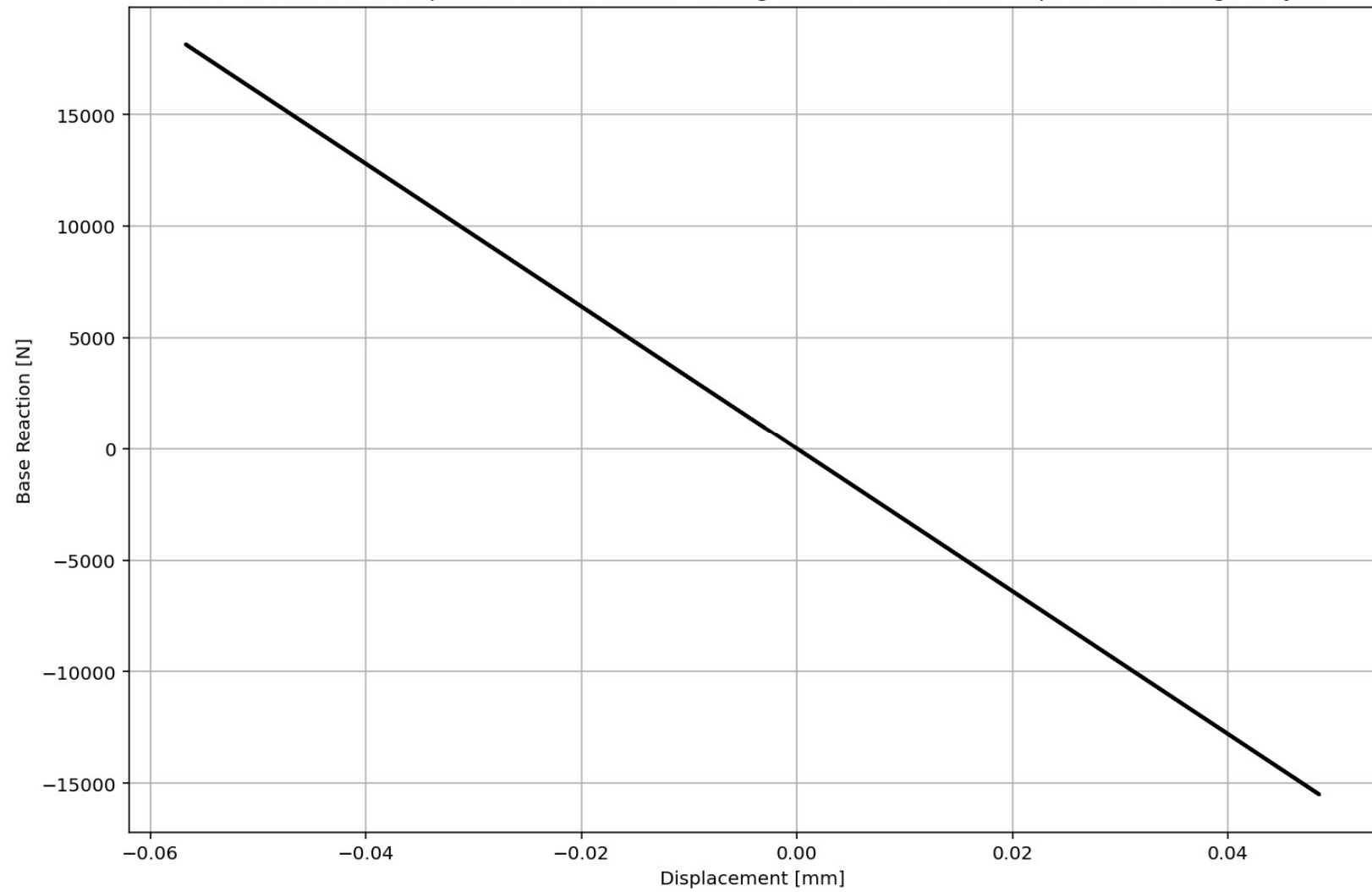
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

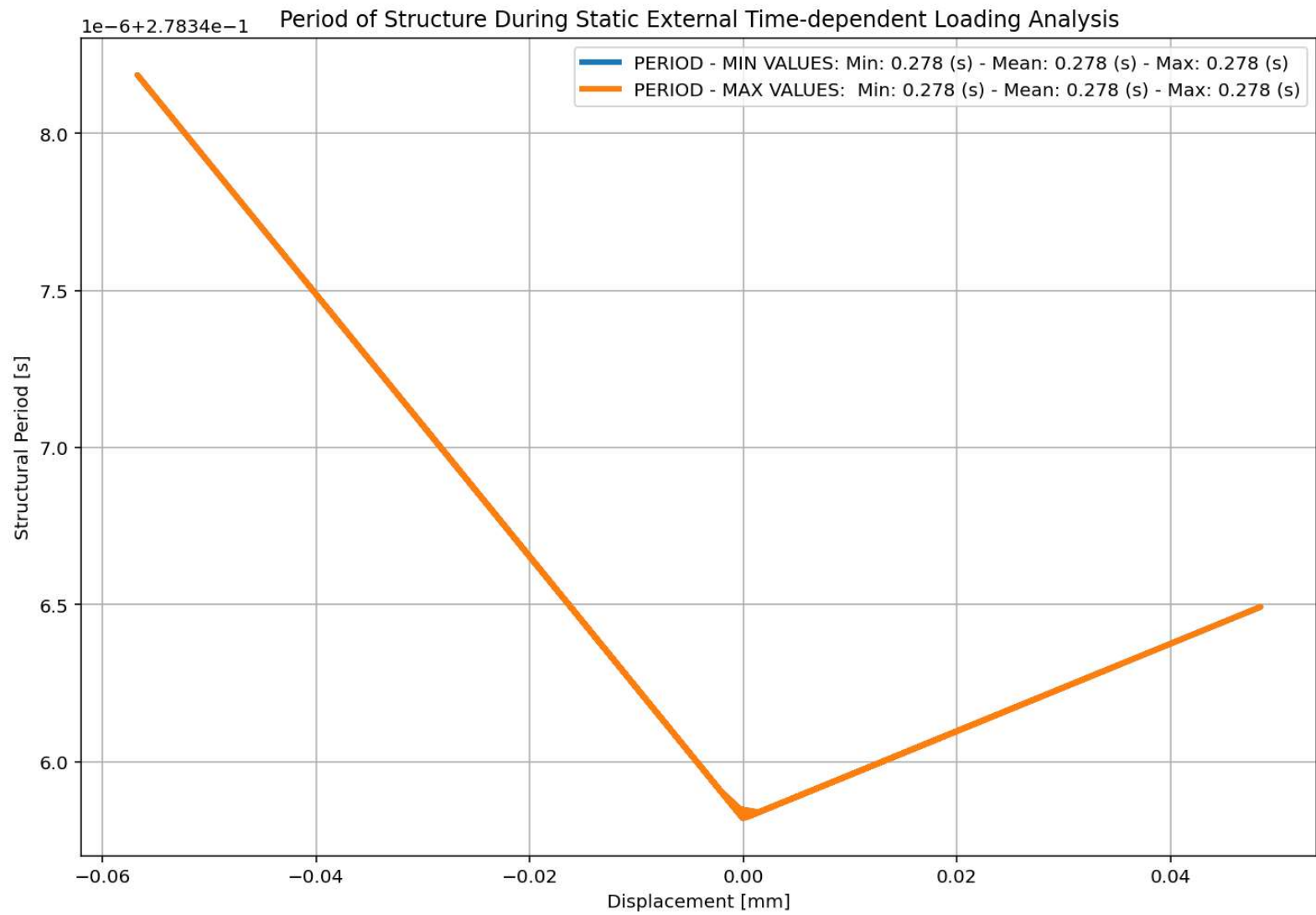
[Python Console] Files Help Variable Explorer Debugger Plots History

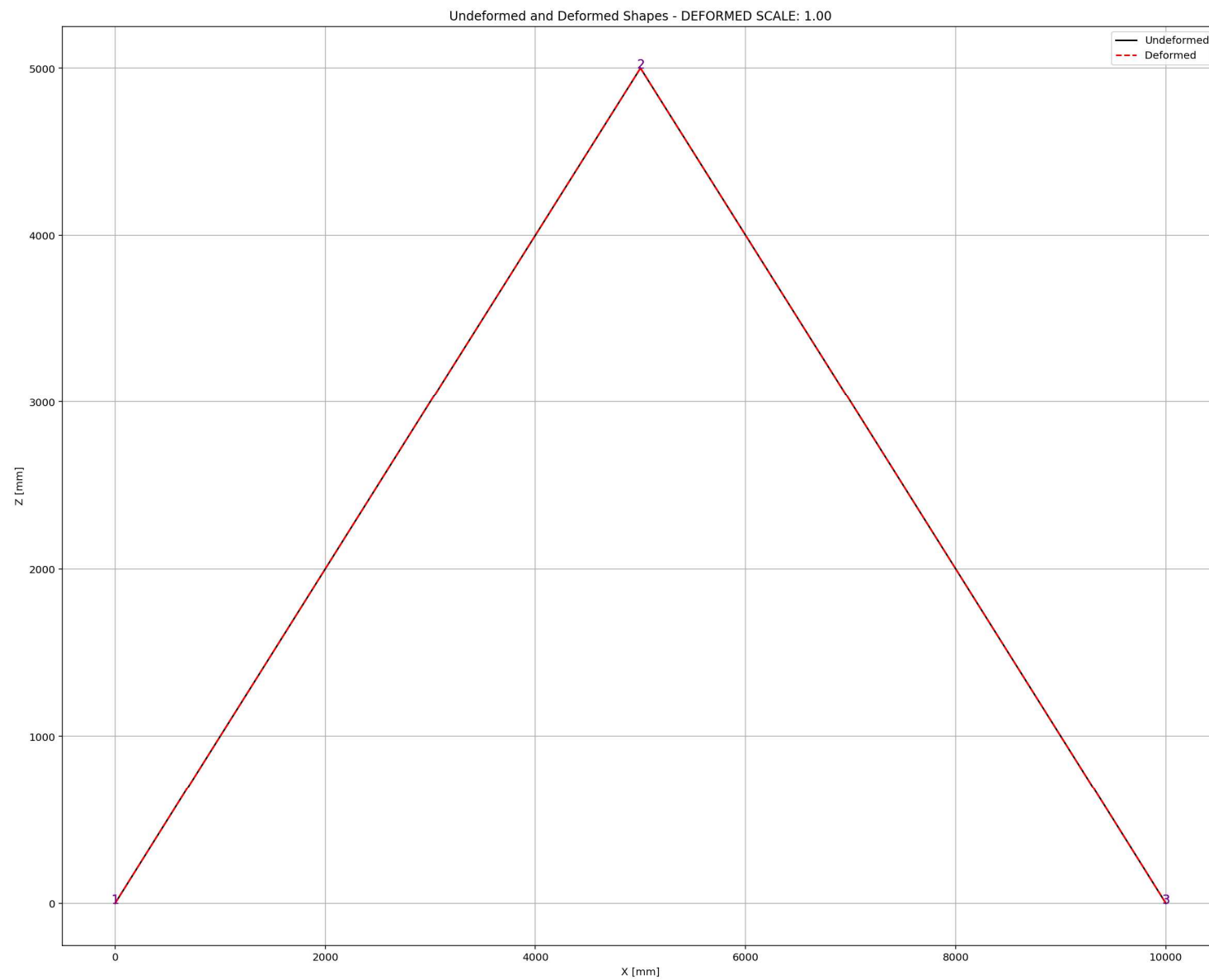
Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 7, Col 44 UTF-8 CRLF RW Mem 50%



Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis







DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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P(t)_TRUSS_TWO_ELEMENTS.py

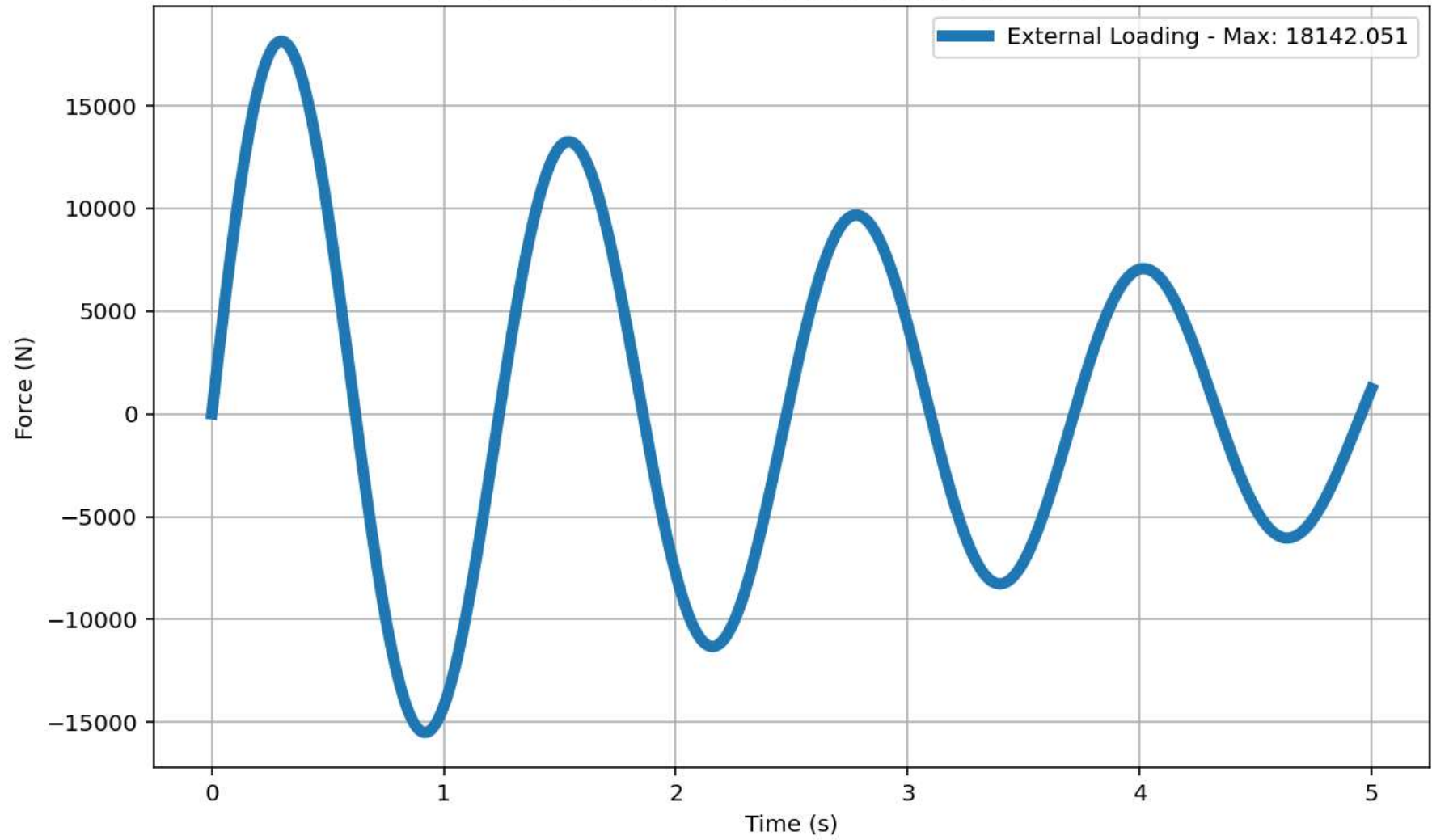
```
518 if ANAL TYPE == 'DYNAMIC EXTERNAL TIME-DEPENDENT LOADING': # DYNAMIC TIME-HISTORY ANAL
519     %% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
520     duration = 20.0           # [s] Analysis duration
521     dt = 0.01                # [s] Time step
522     DT = dt                  # [s] Time step
523     DT_time = 5.0            # [s] Total external Load Analysis Durations [*****]
524     force_amplitude = 19600.0 # [N] Amplitude Force
525     omega_DT = 5.0715        # [rad/s] Natural angular frequency
526     DR = 0.03                # Damping Ratio
527
528     # Check function
529     def CHECK_FUN(DT_time ,duration):
530         if DT_time > duration:
531             print('\n\nAnalysis Duration Must be greater than External Load Duration\n\n')
532             exit()
533         return -1
534
535     CHECK_FUN(DT_time ,duration)
536     def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time): # P(t) = P0 sin
537         import numpy as np
538         import matplotlib.pyplot as plt
539         # External Load Durations
540         num_steps = int(DT_time / DT)
541         load_time = np.linspace(0, DT_time, num_steps)
542         target_frequency = 1.0 * omega_DT # Target excitation frequency
543         DT_load = force_amplitude * np.sin(target_frequency * load_time)
544         # Plot External Time-dependent Loading
545         plt.figure(figsize=(10, 6))
546         plt.plot(load_time, DT_load, label=f'External Loading - Max: {np.max(DT_load)}:
547         plt.title('External Time-dependent Loading Over Time')
548         plt.xlabel('Time (s)')
549         plt.ylabel('Force (N)')
550         plt.grid(True)
551         plt.legend()
```

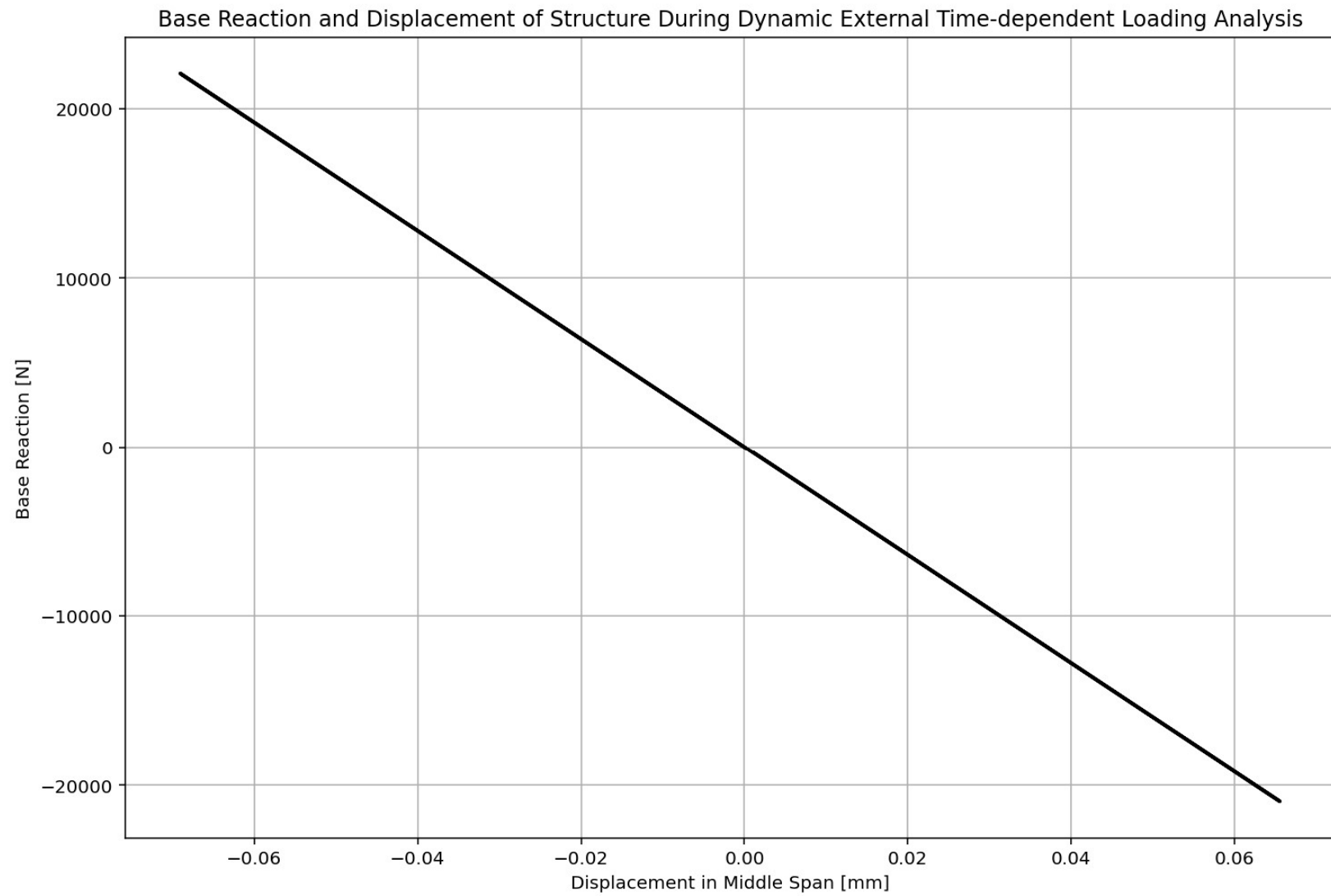
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis

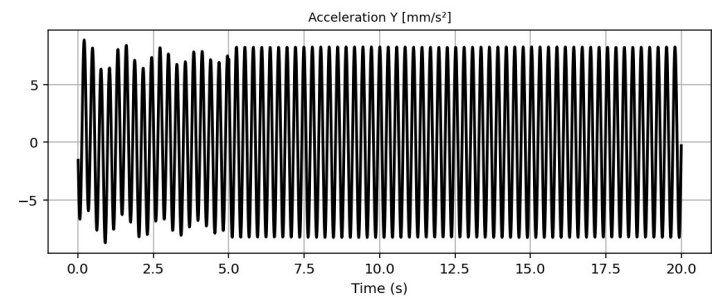
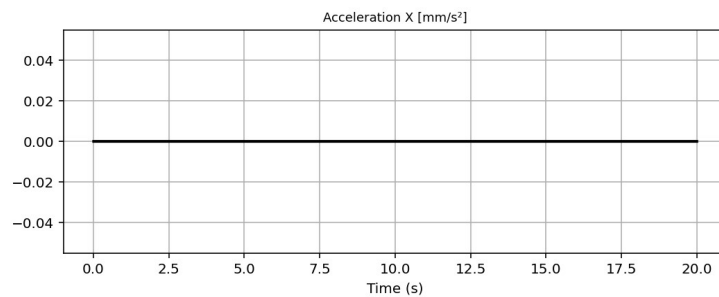
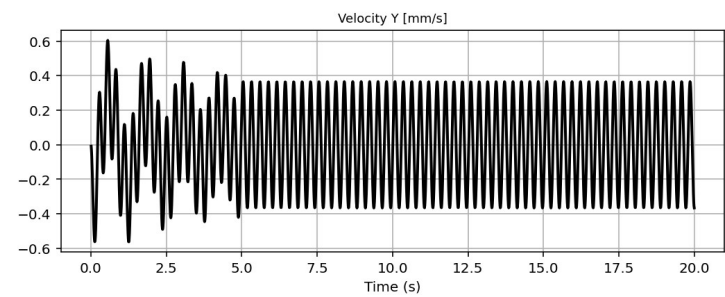
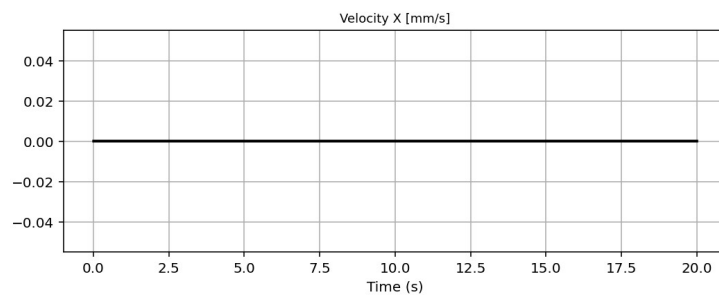
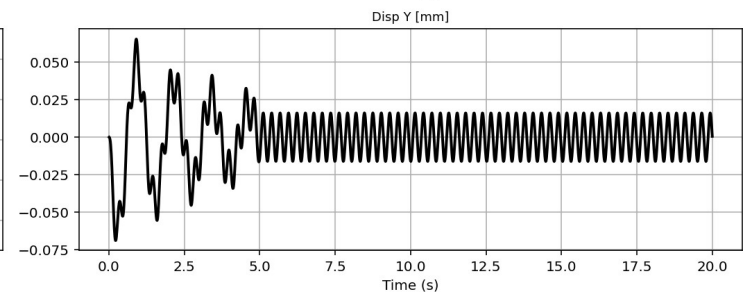
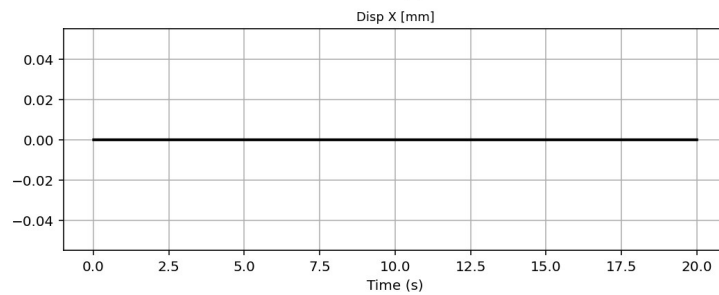
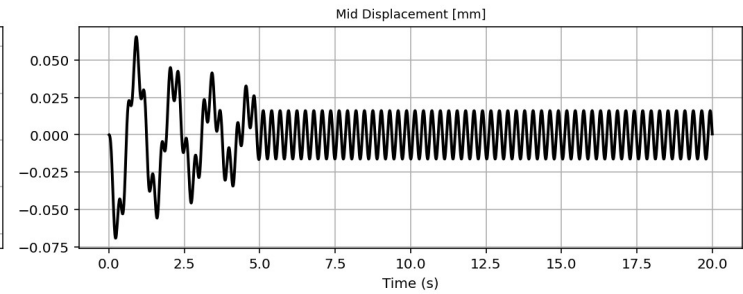
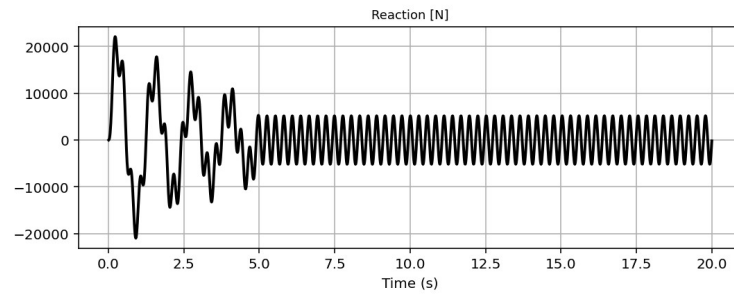
IPython Console Files Help Variable Explorer Debugger Plots History

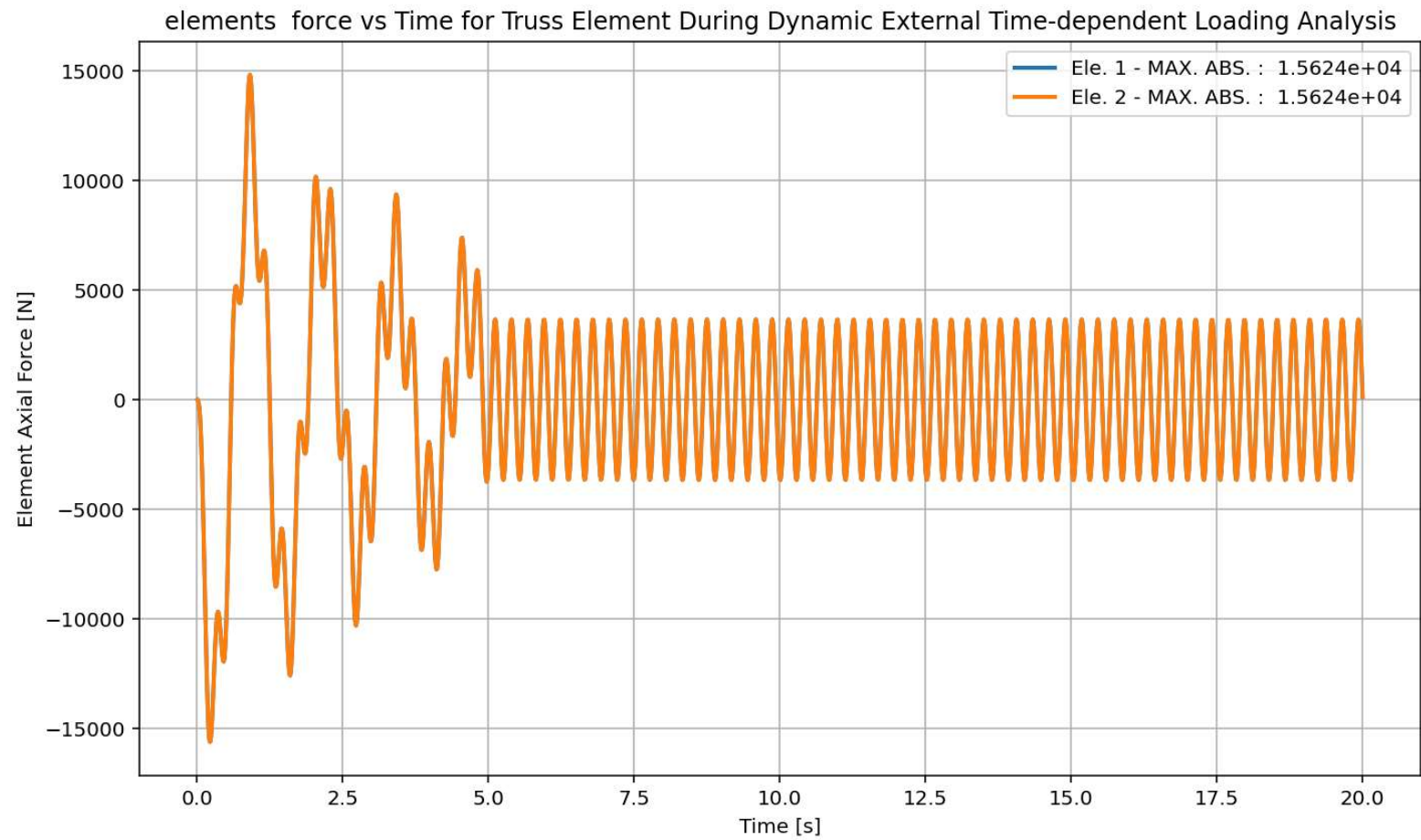
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External Time-dependent Loading Over Time

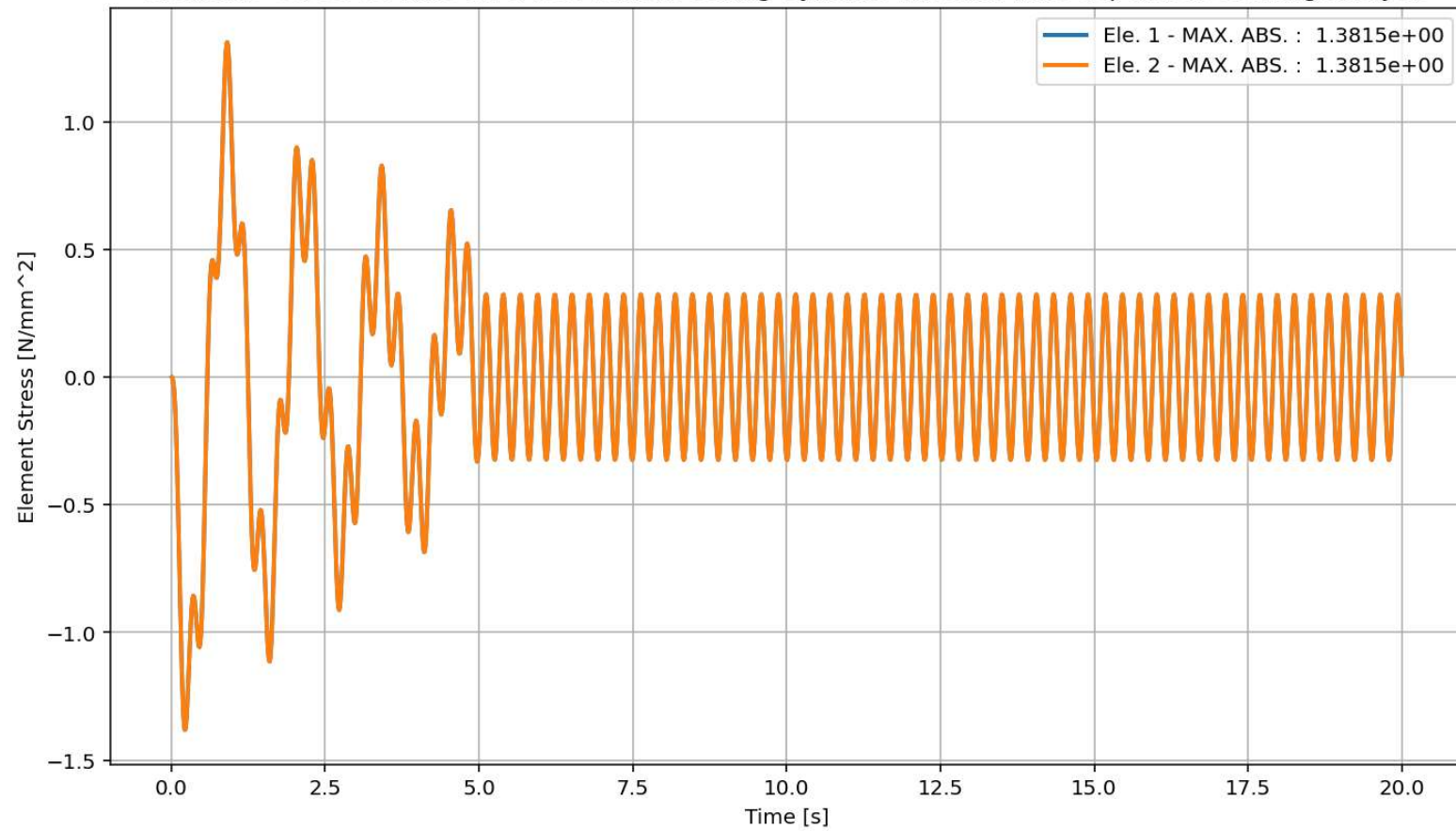




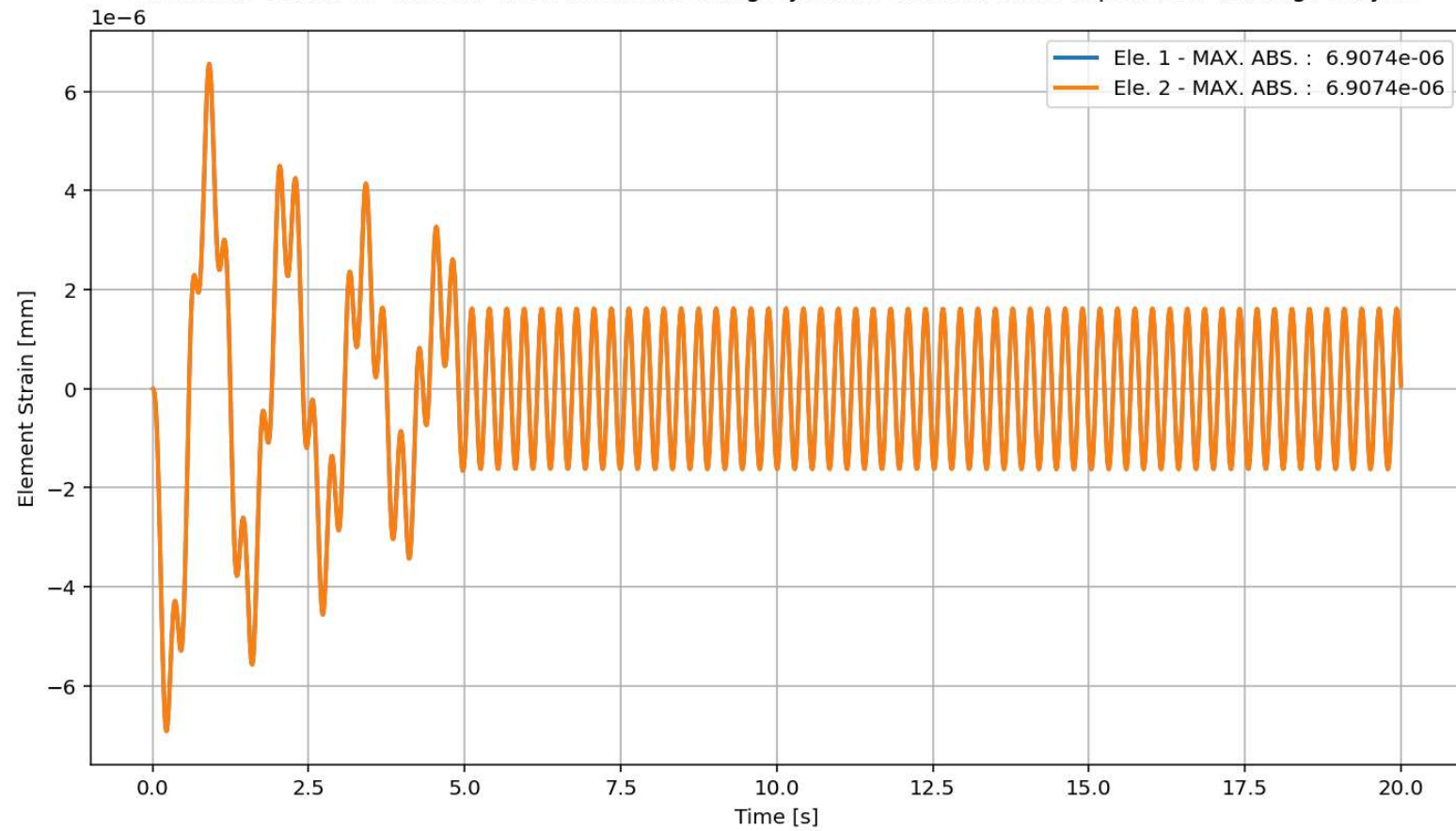




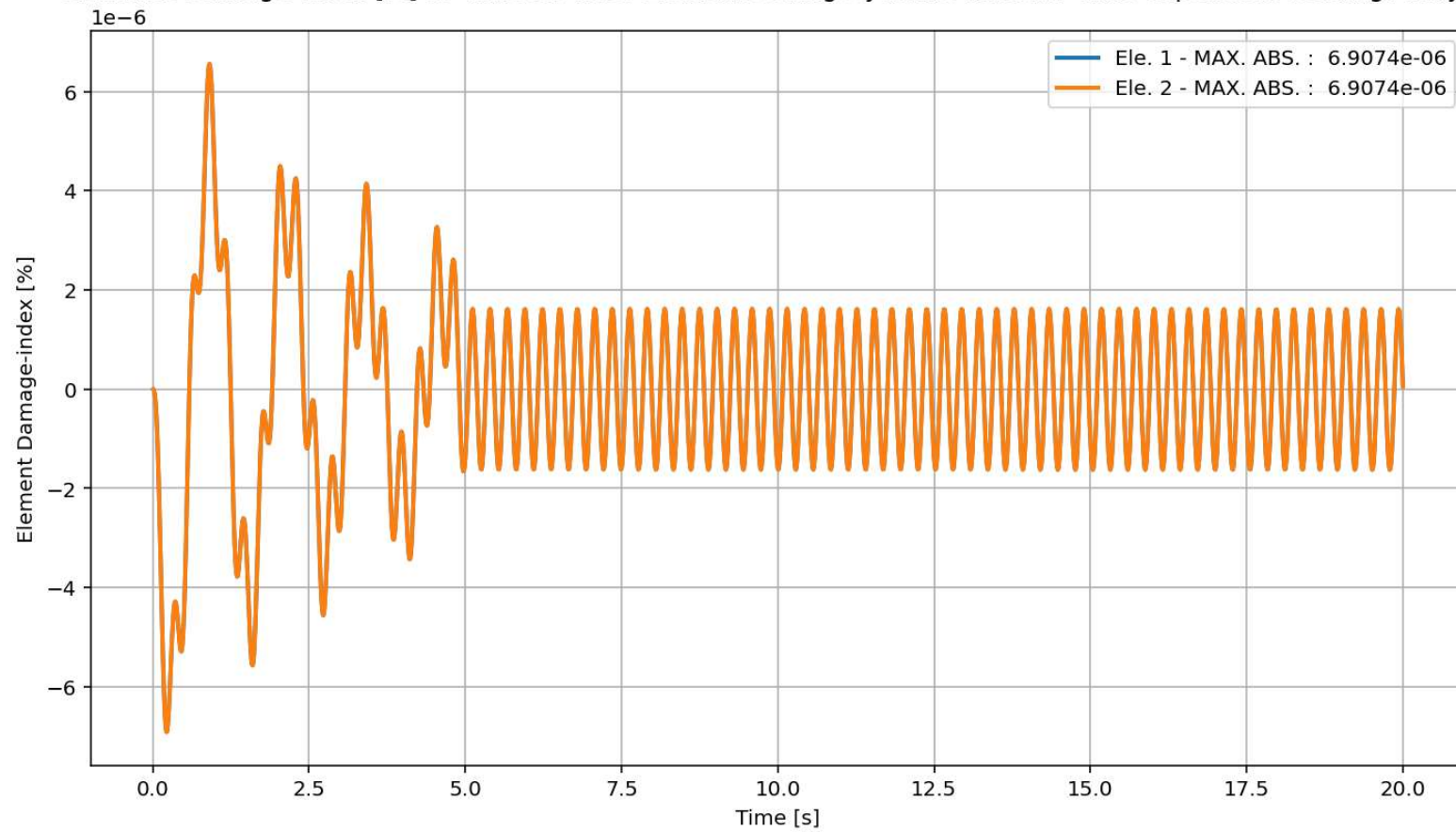
elements Stress vs Time for Truss Element During Dynamic External Time-dependent Loading Analysis



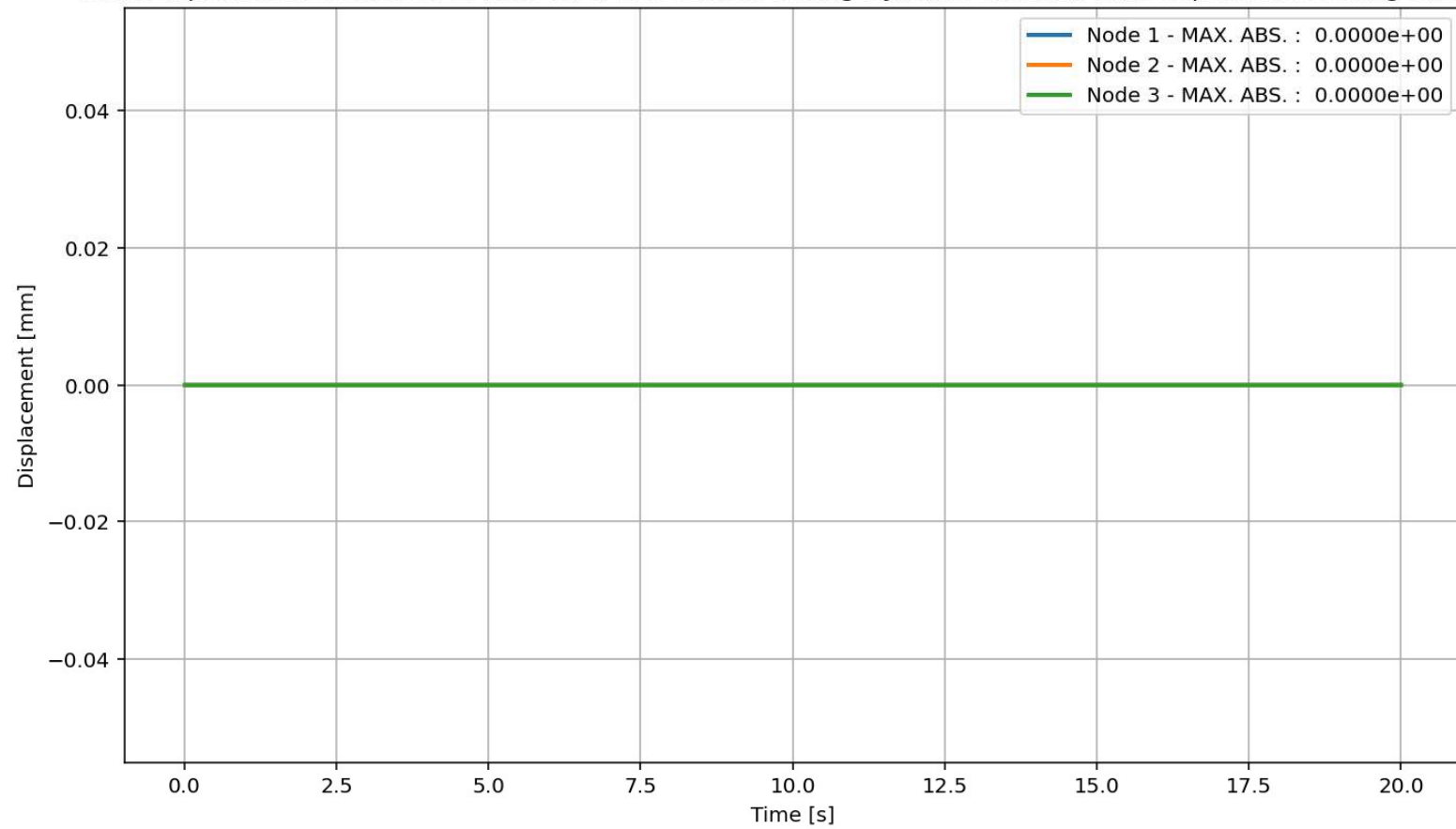
elements Strain vs Time for Truss Element During Dynamic External Time-dependent Loading Analysis



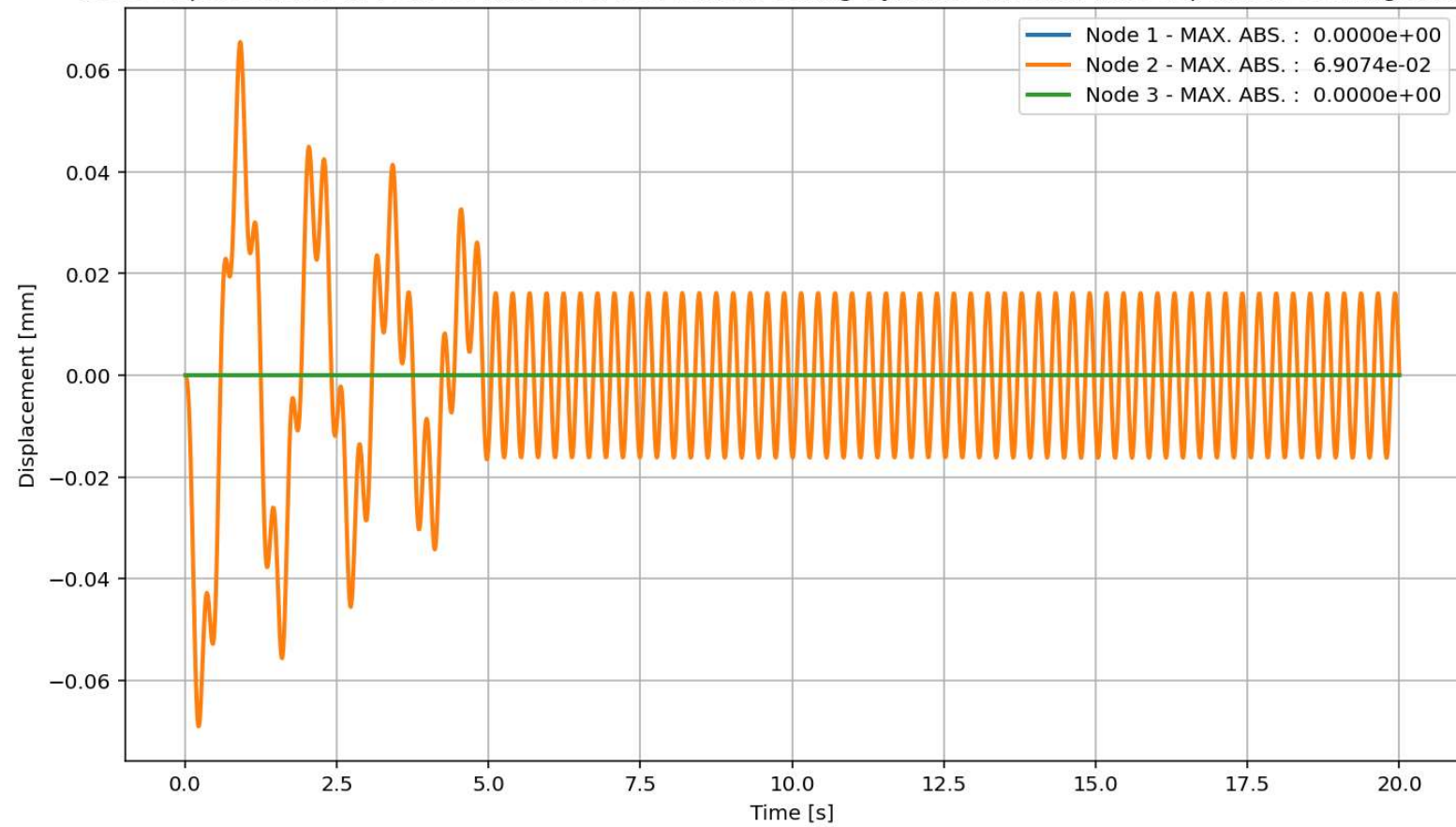
elements Damage-index [%] vs Time for Truss Element During Dynamic External Time-dependent Loading Analysis

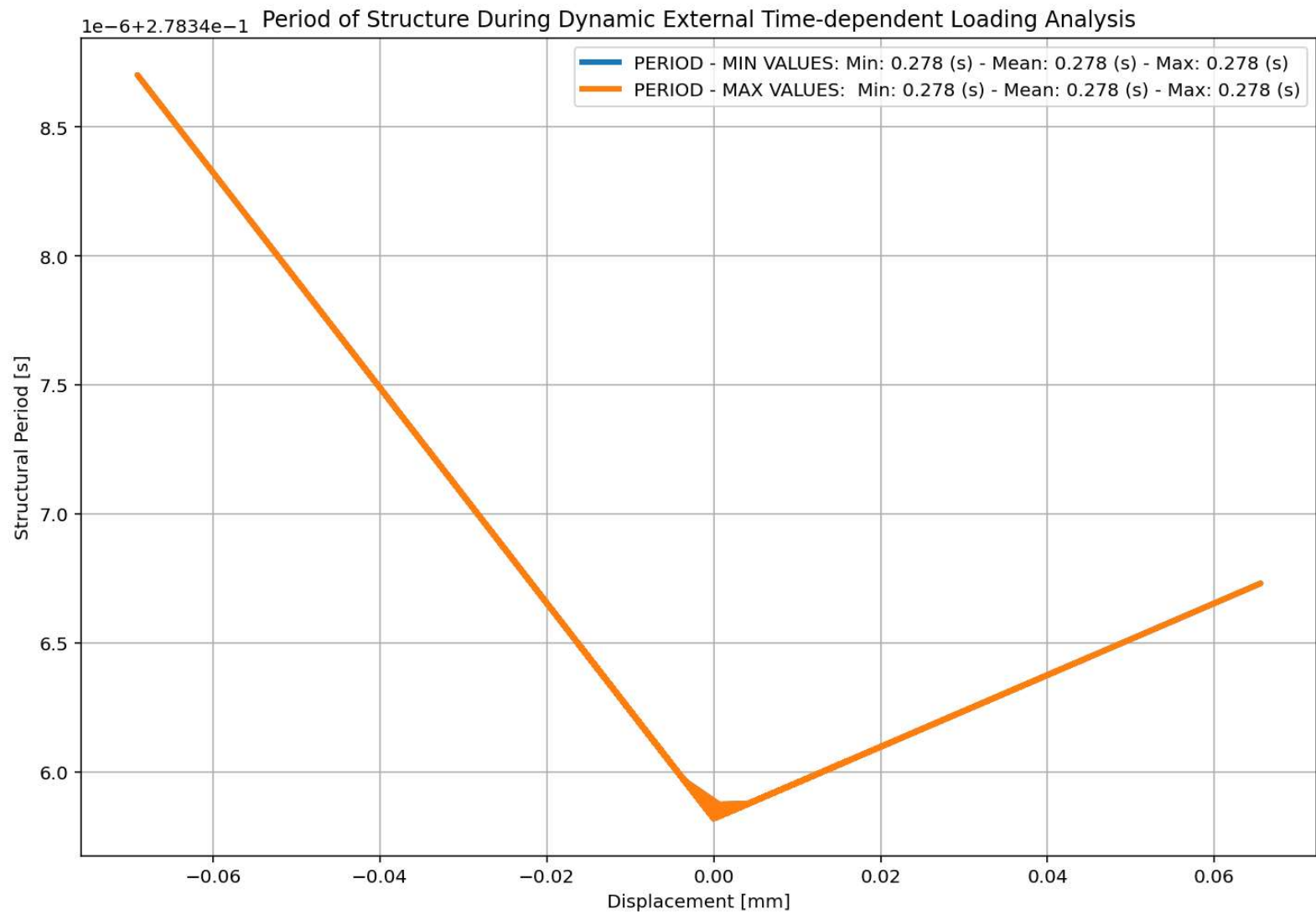


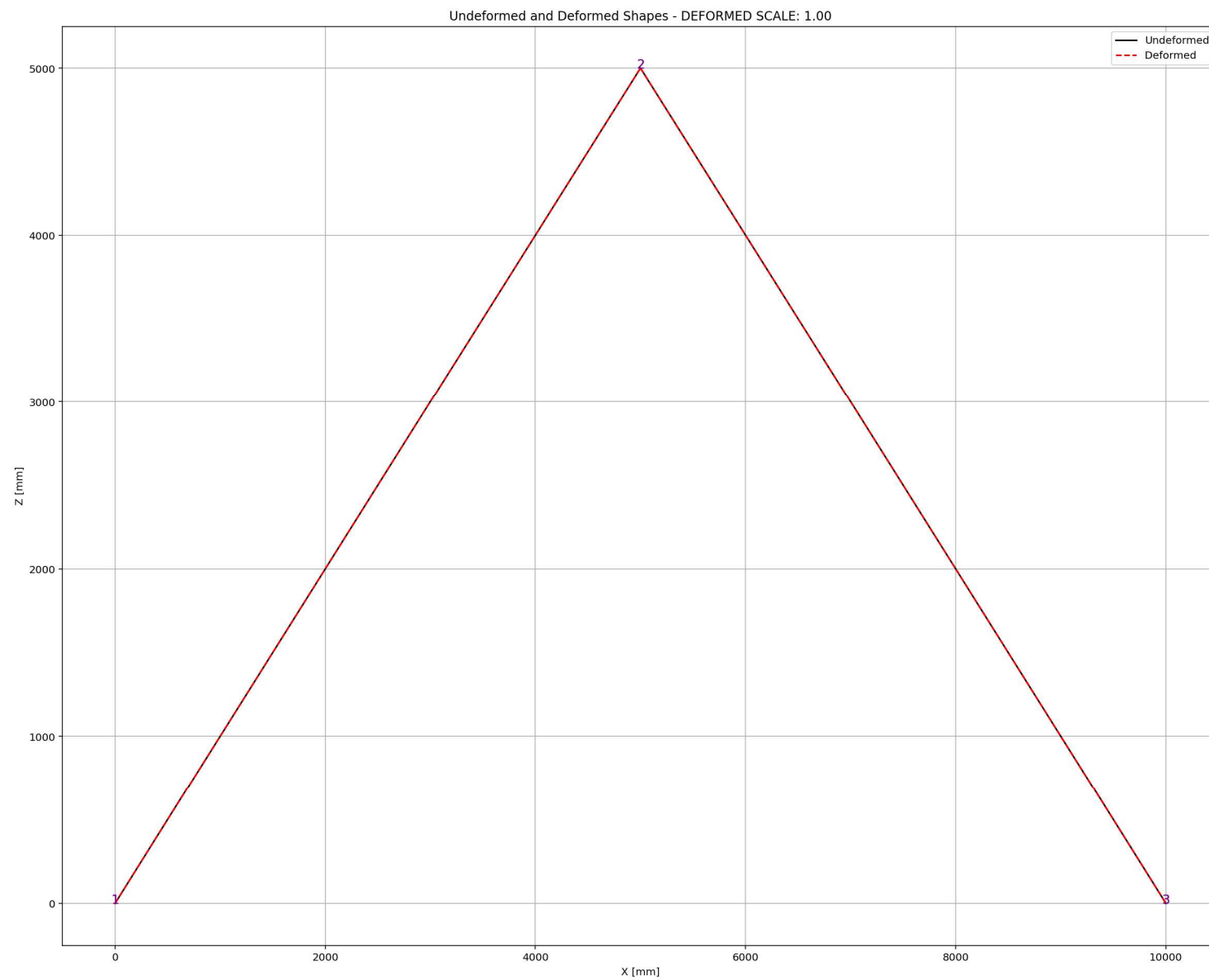
Node Displacements in X Dir. vs Time for Truss Element During Dynamic External Time-dependent Loading Analysis



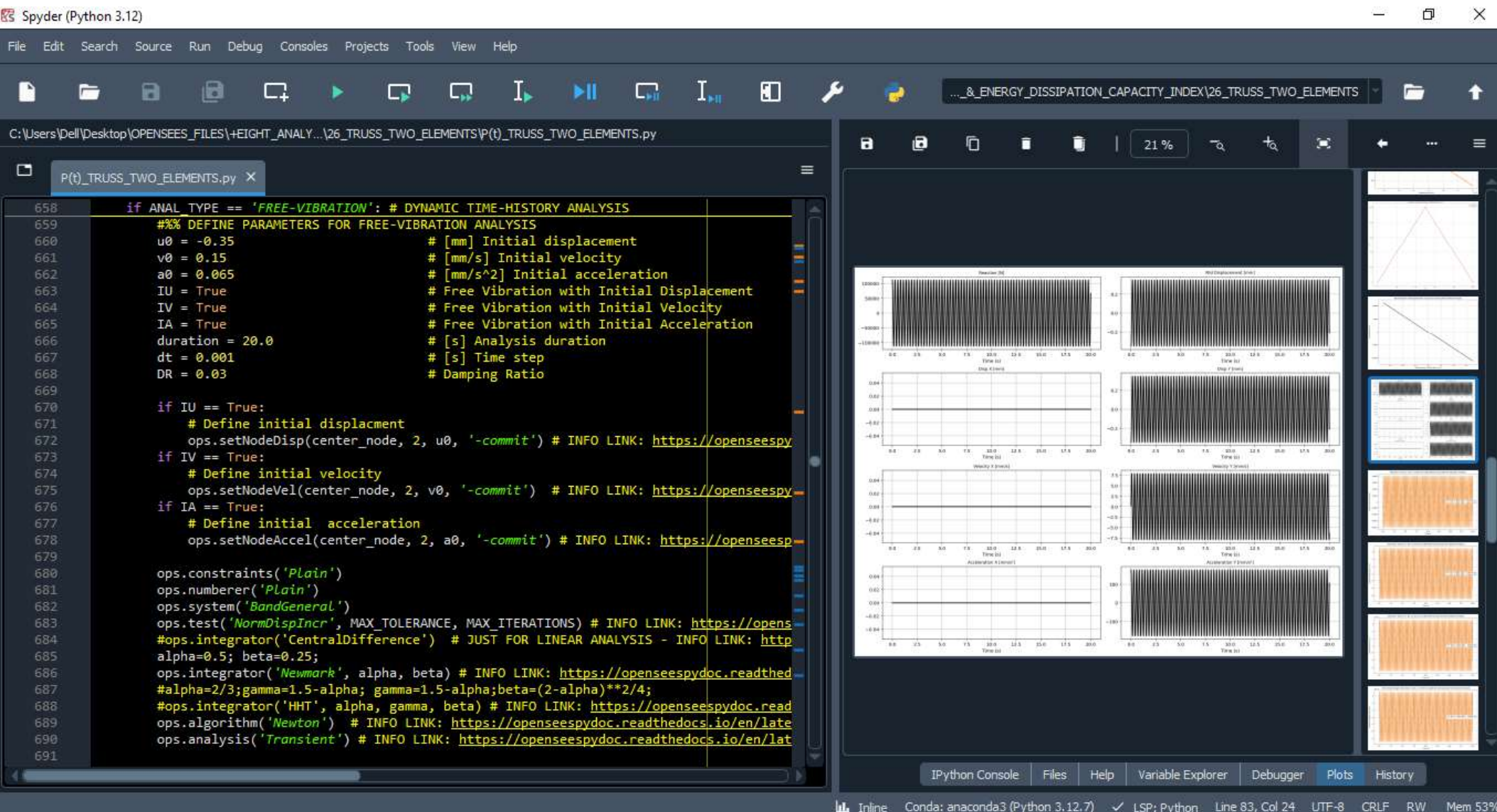
Node Displacements in Y Dir. vs Time for Truss Element During Dynamic External Time-dependent Loading Analysis

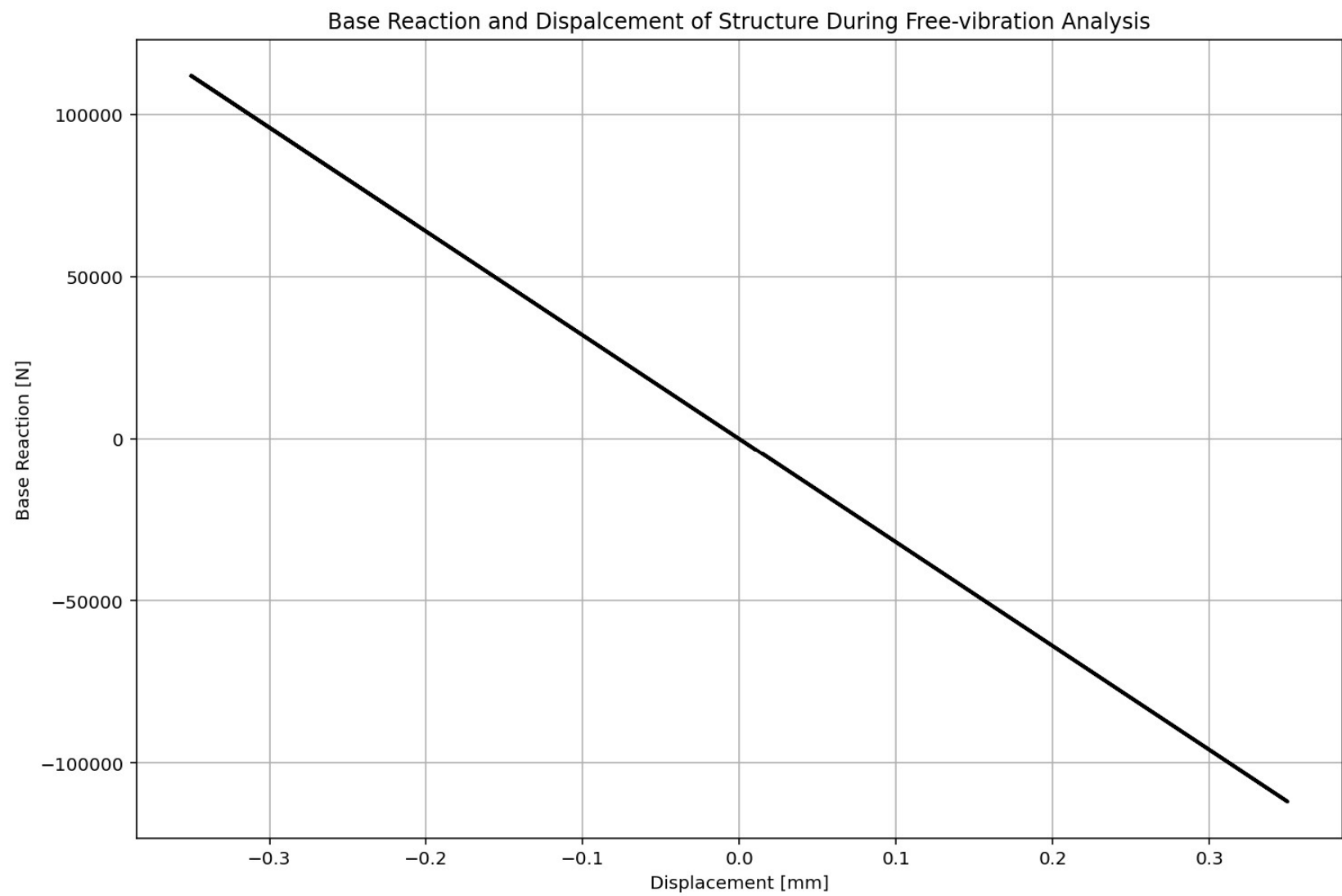


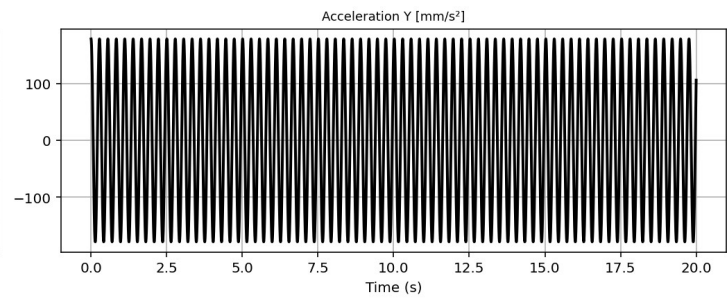
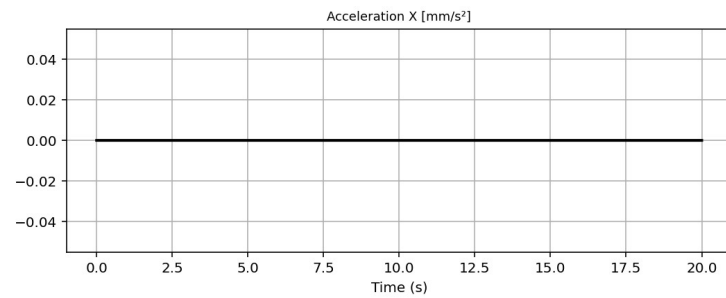
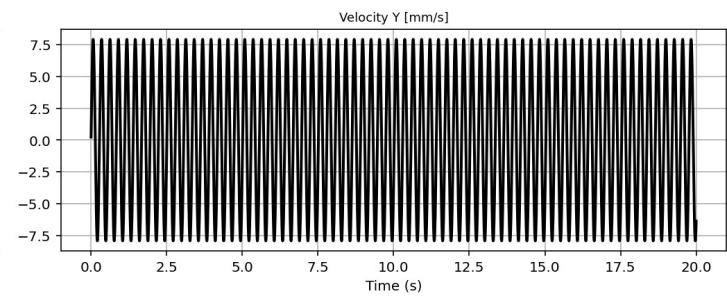
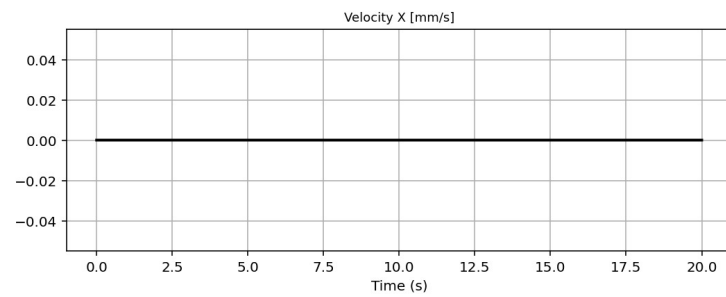
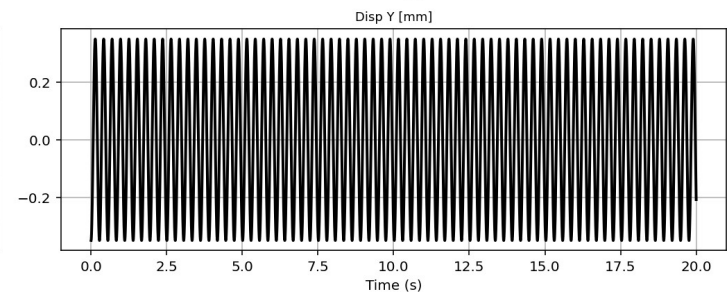
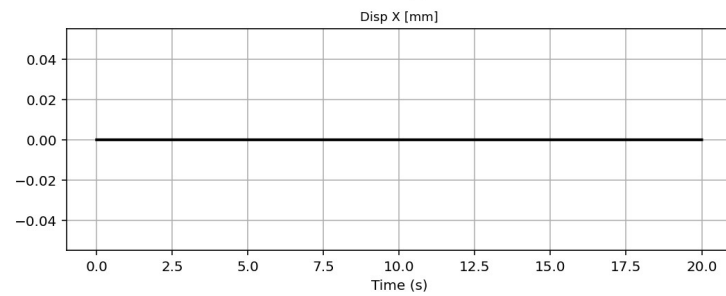
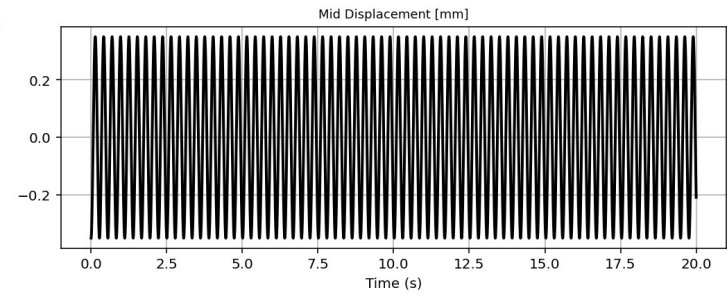
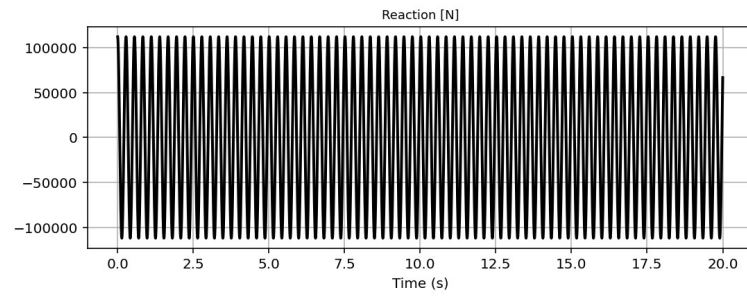


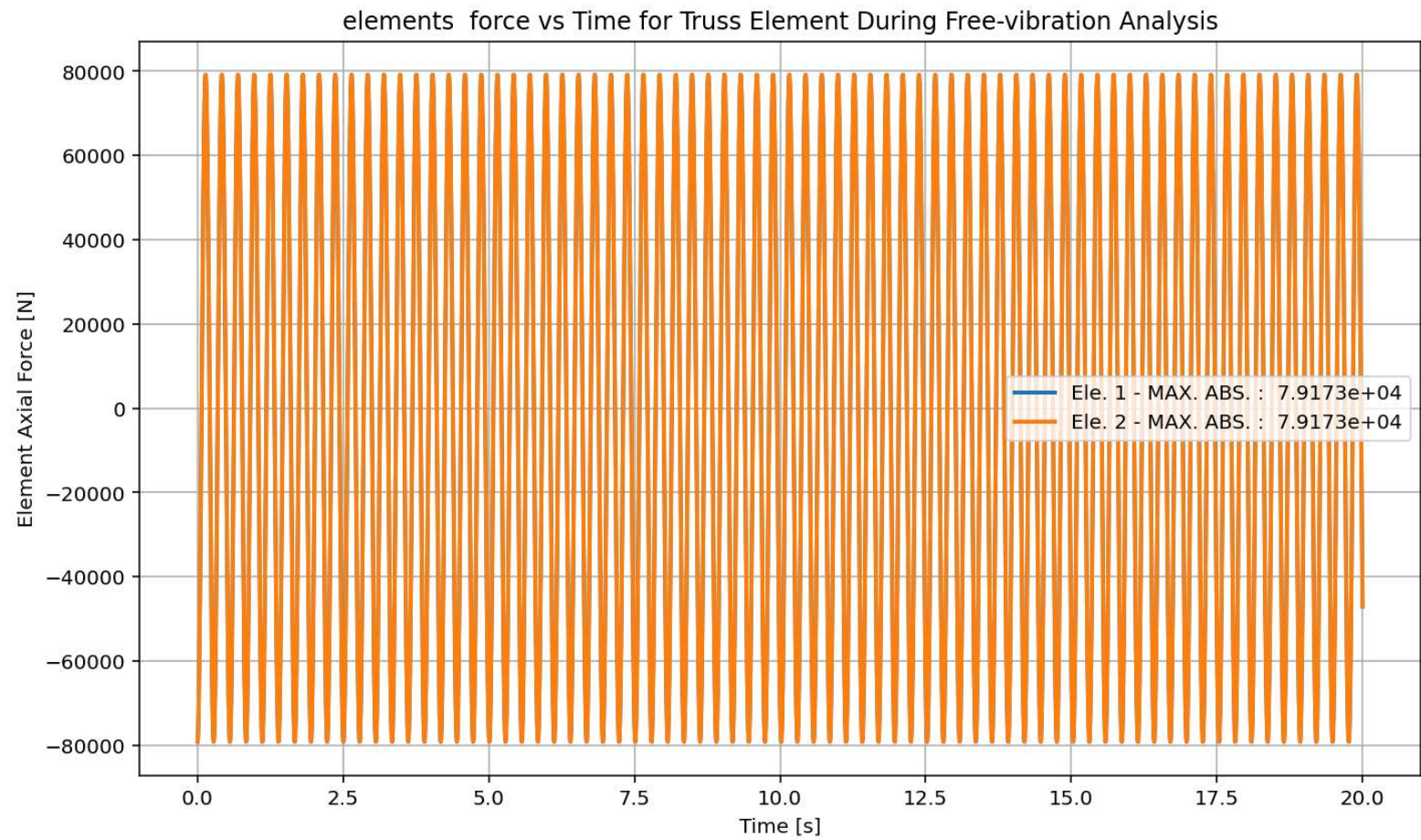


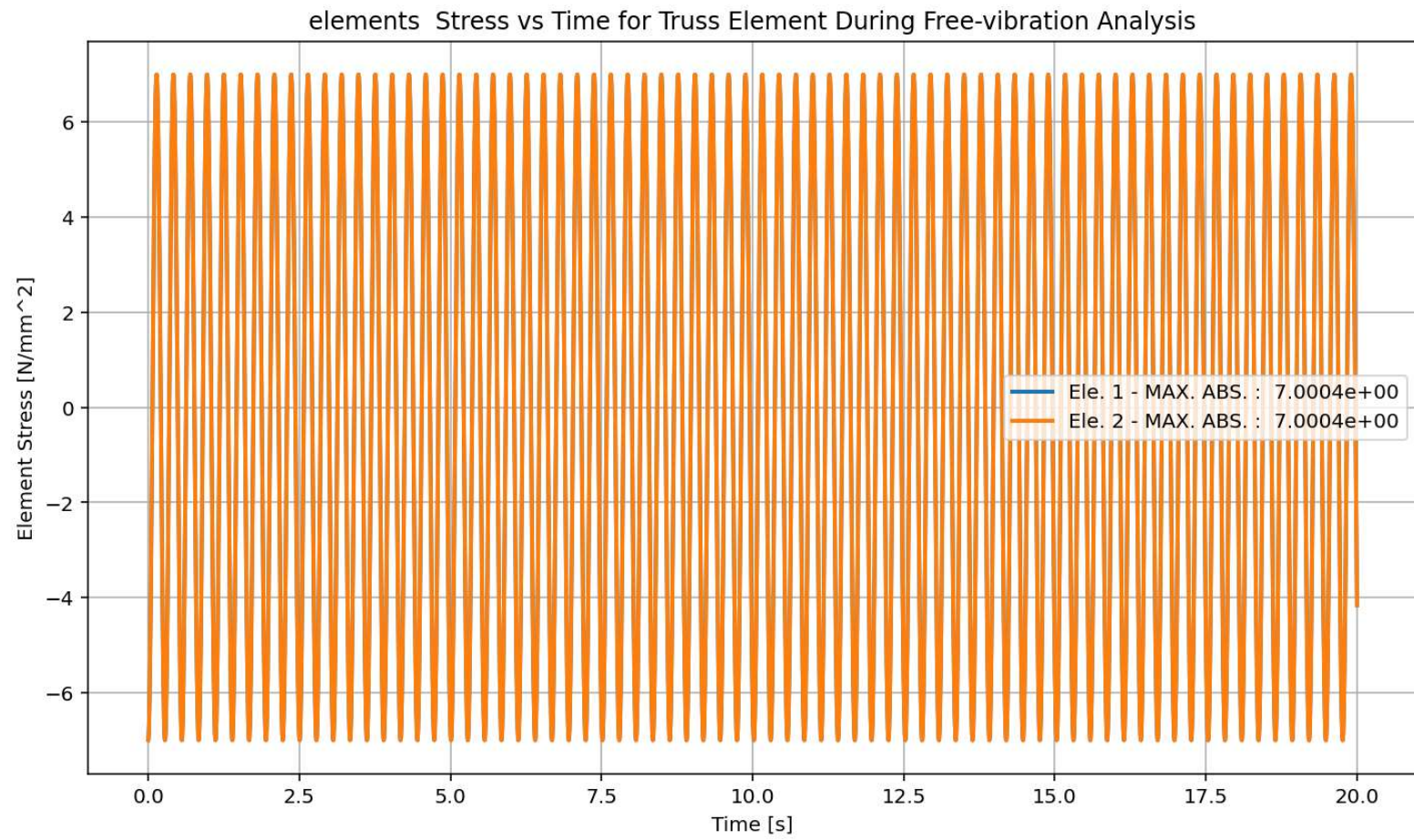
FREE-VIBRATION ANALYSIS



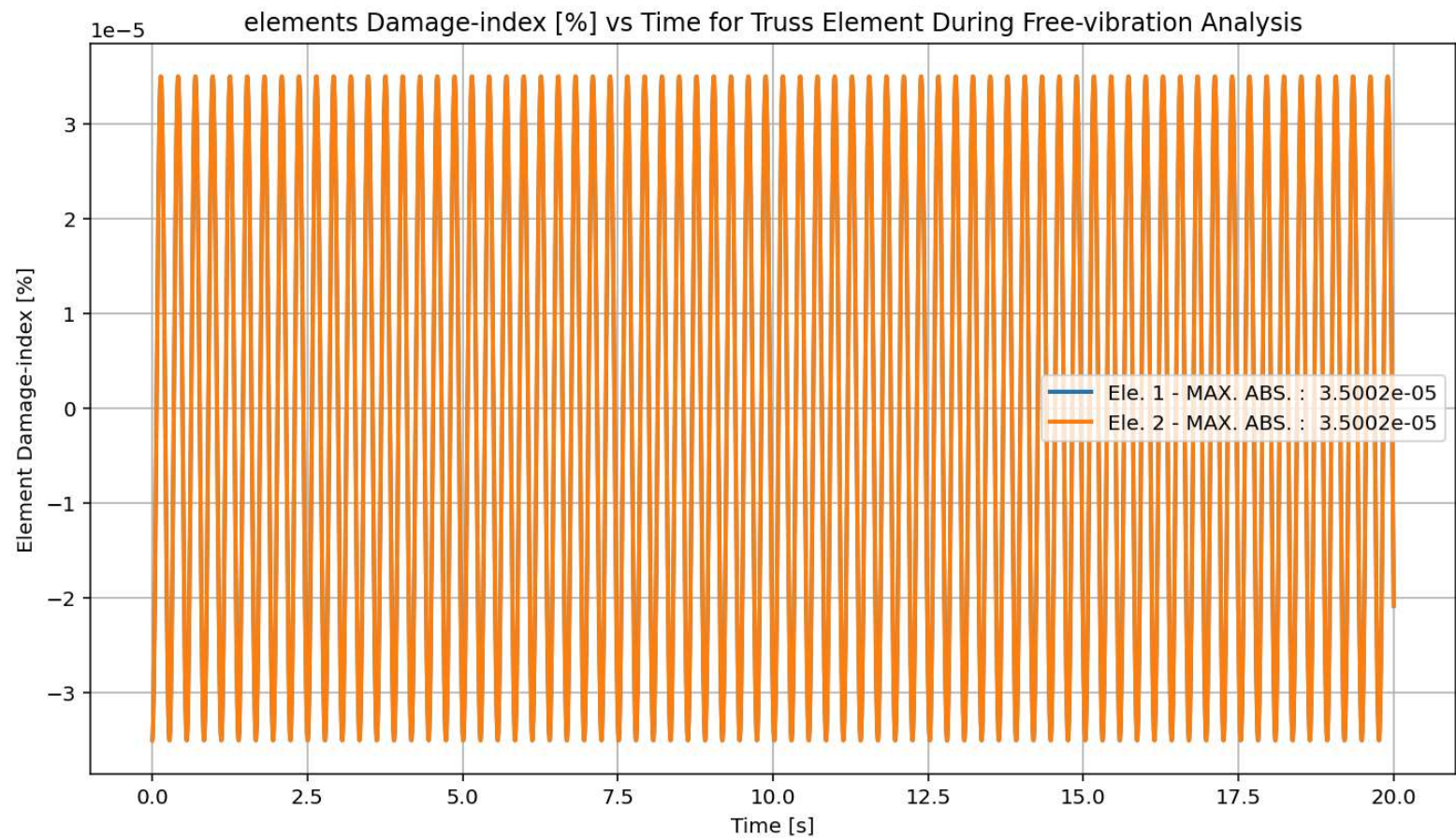


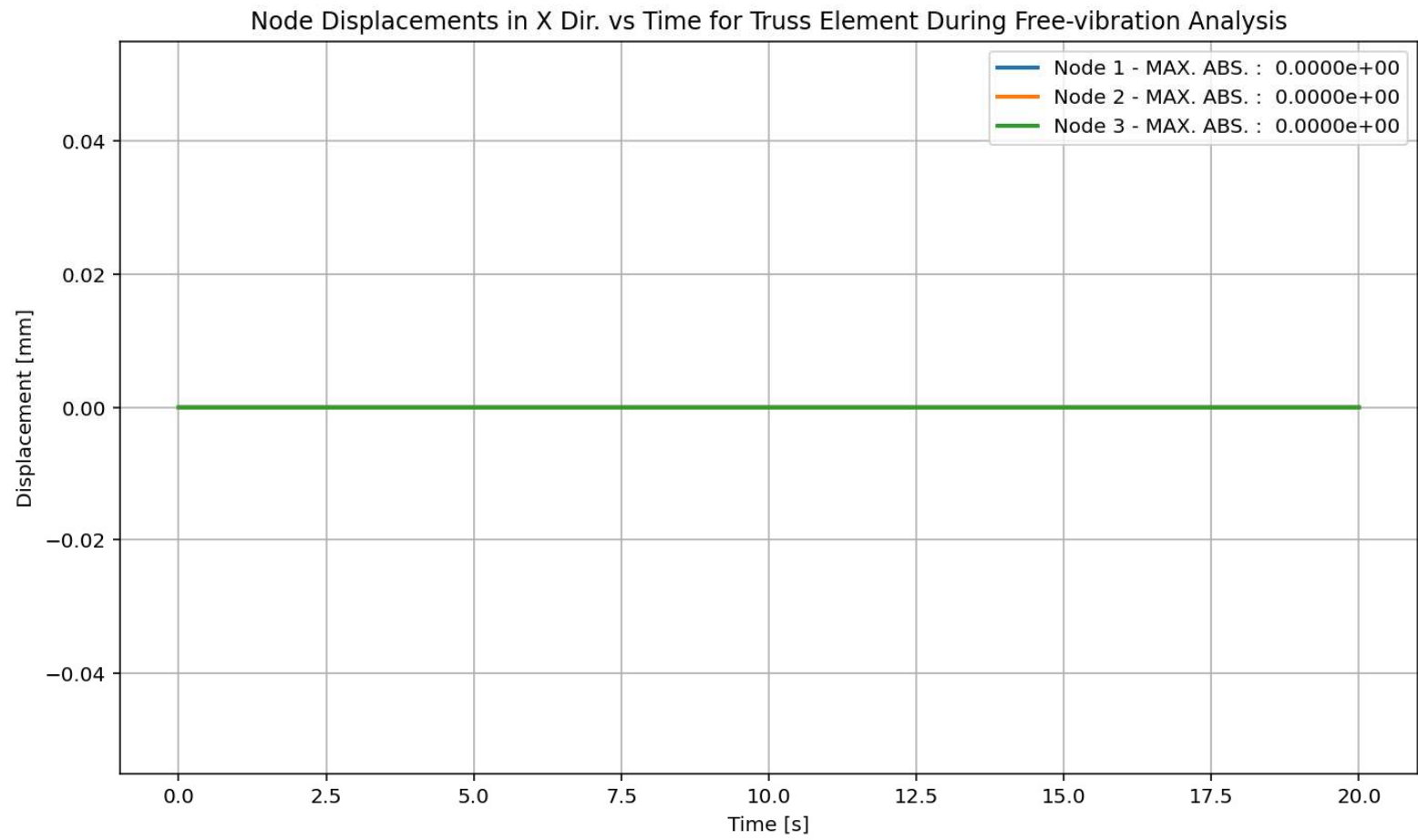


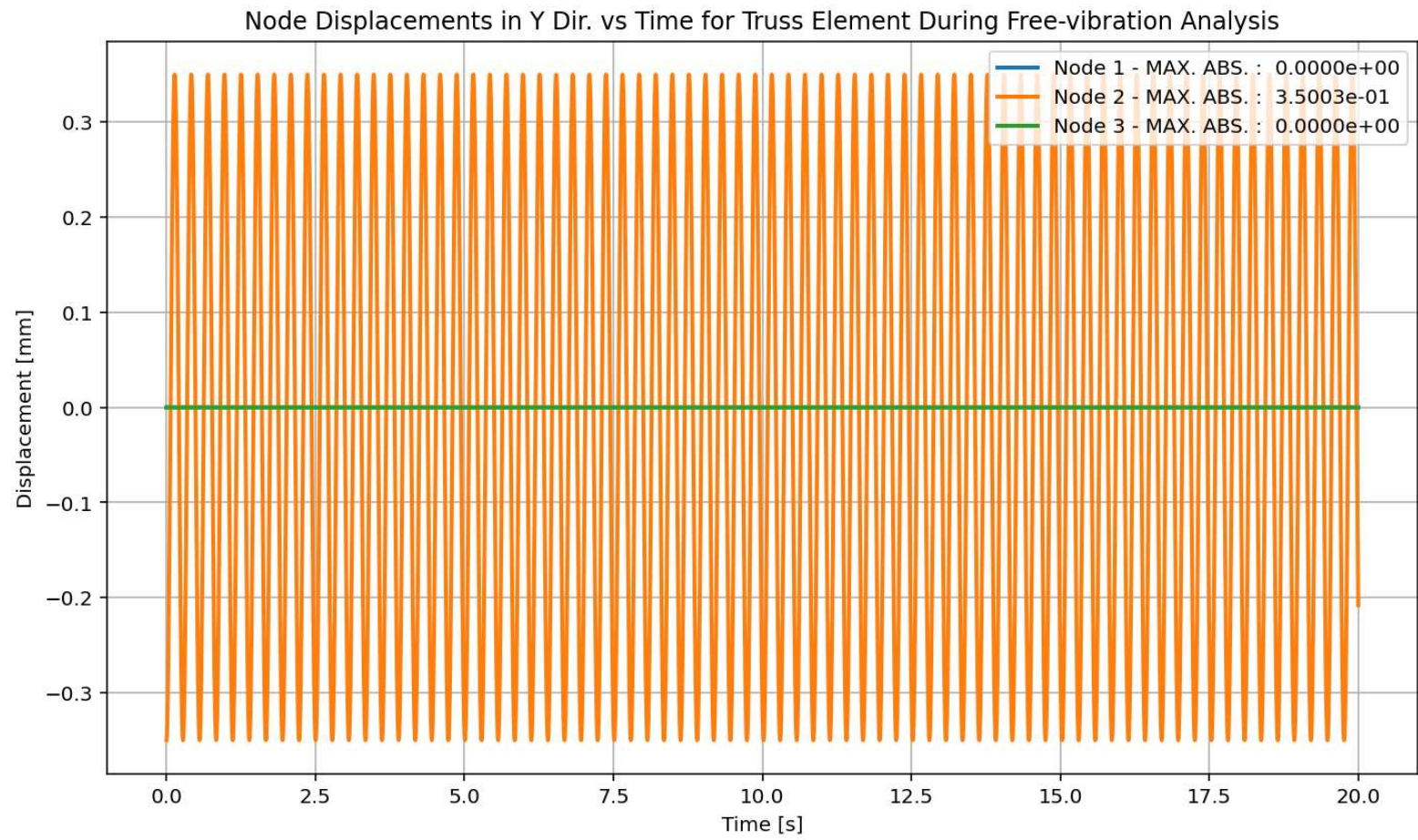


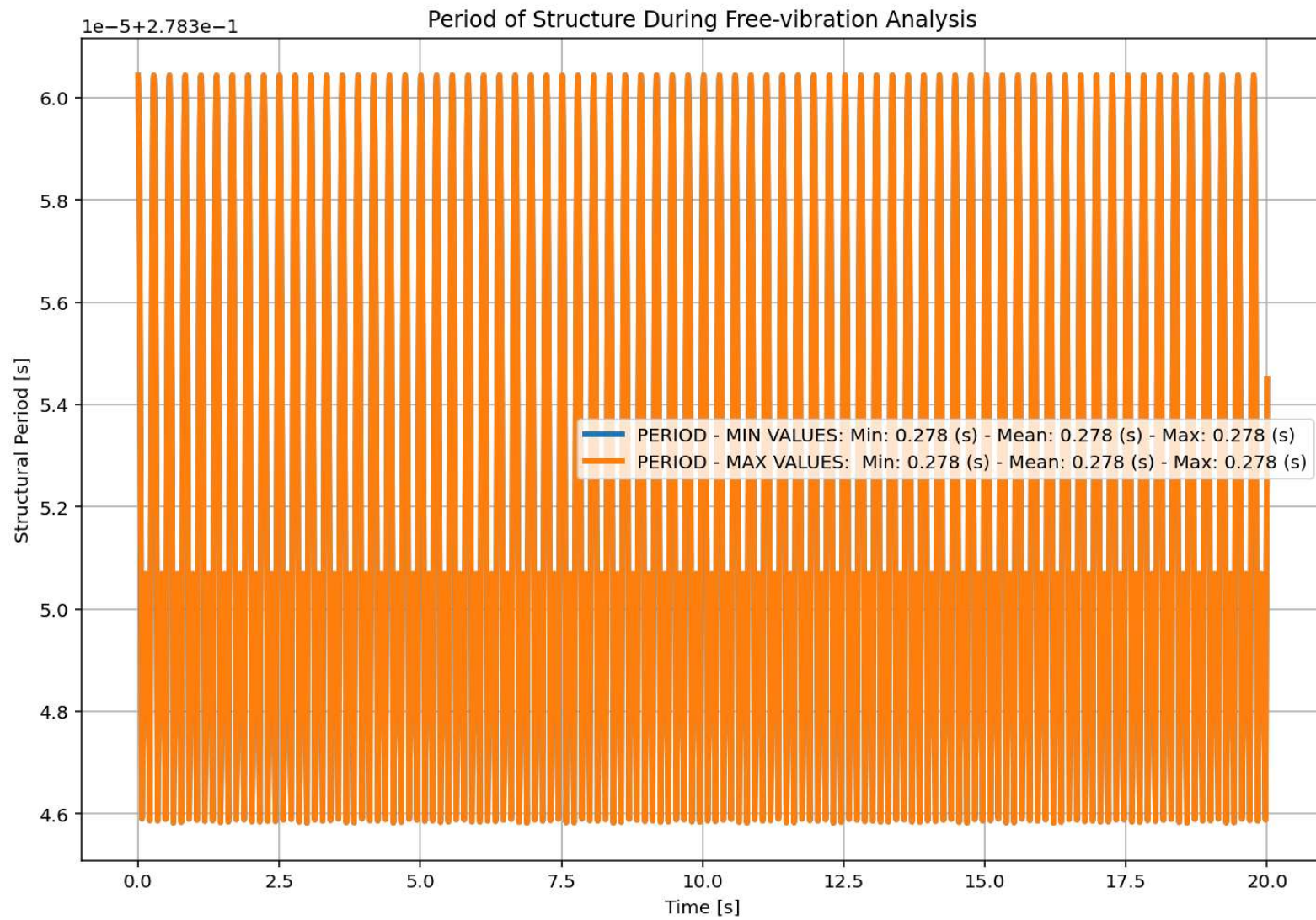




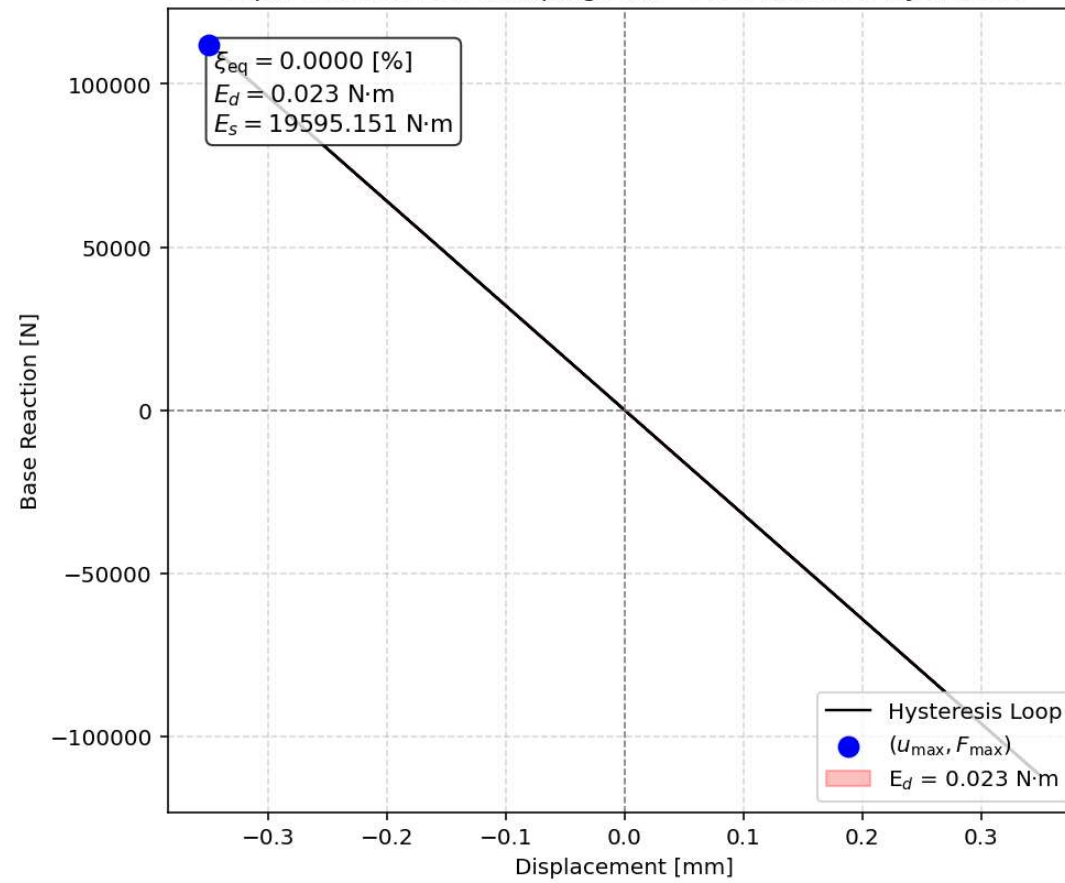


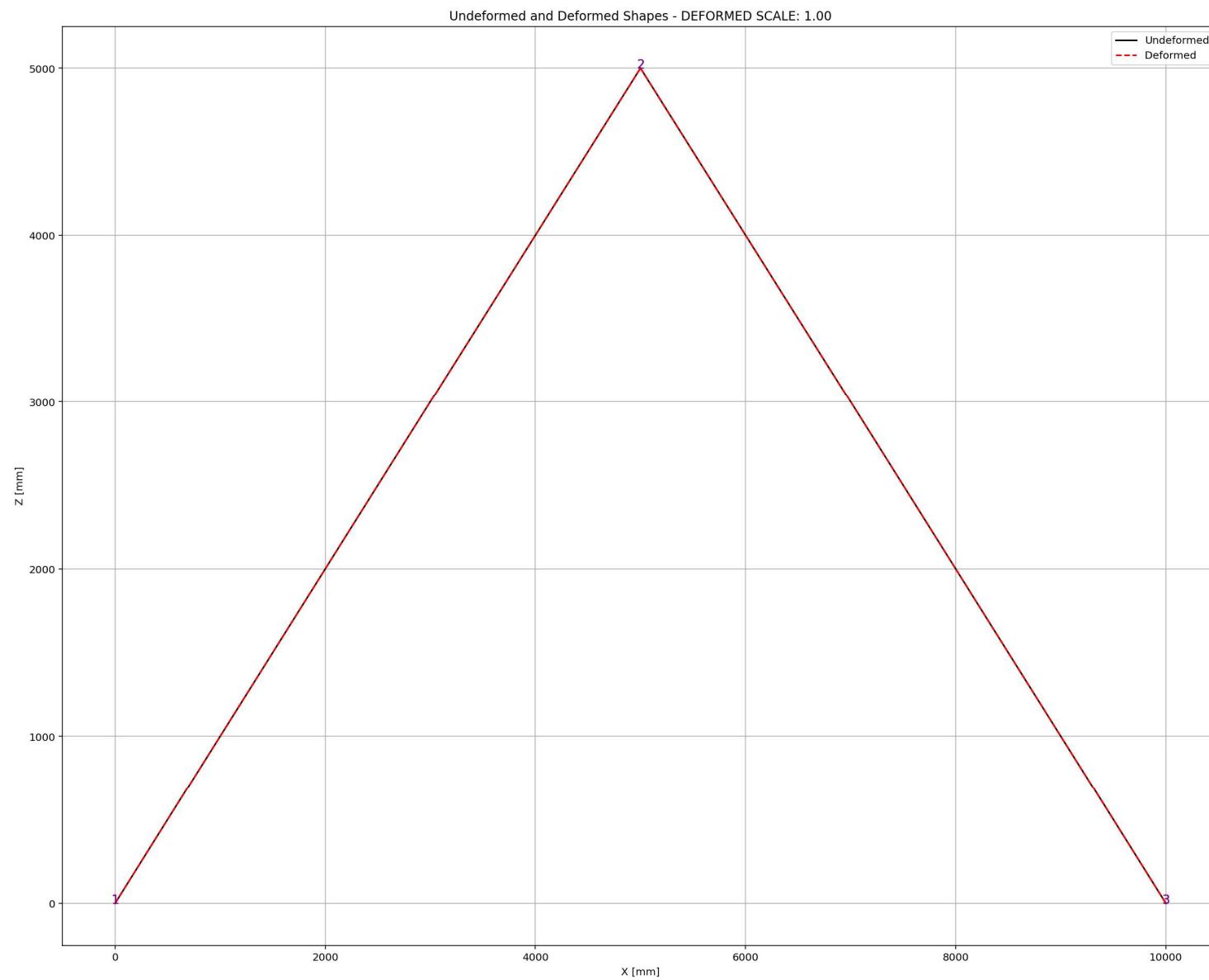






Equivalent viscous damping ratio - Free-vibration Hysteresis





SEISMIC ANALYSIS

Spdyer (Python 3.12)

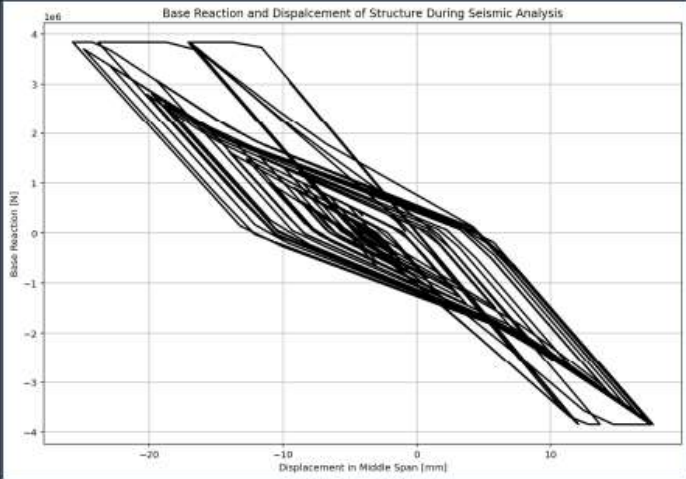
File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...\26_TRUSS_TWO_ELEMENTS\P(t)_TRUSS_TWO_ELEMENTS.py

P(t)_TRUSS_TWO_ELEMENTS.py

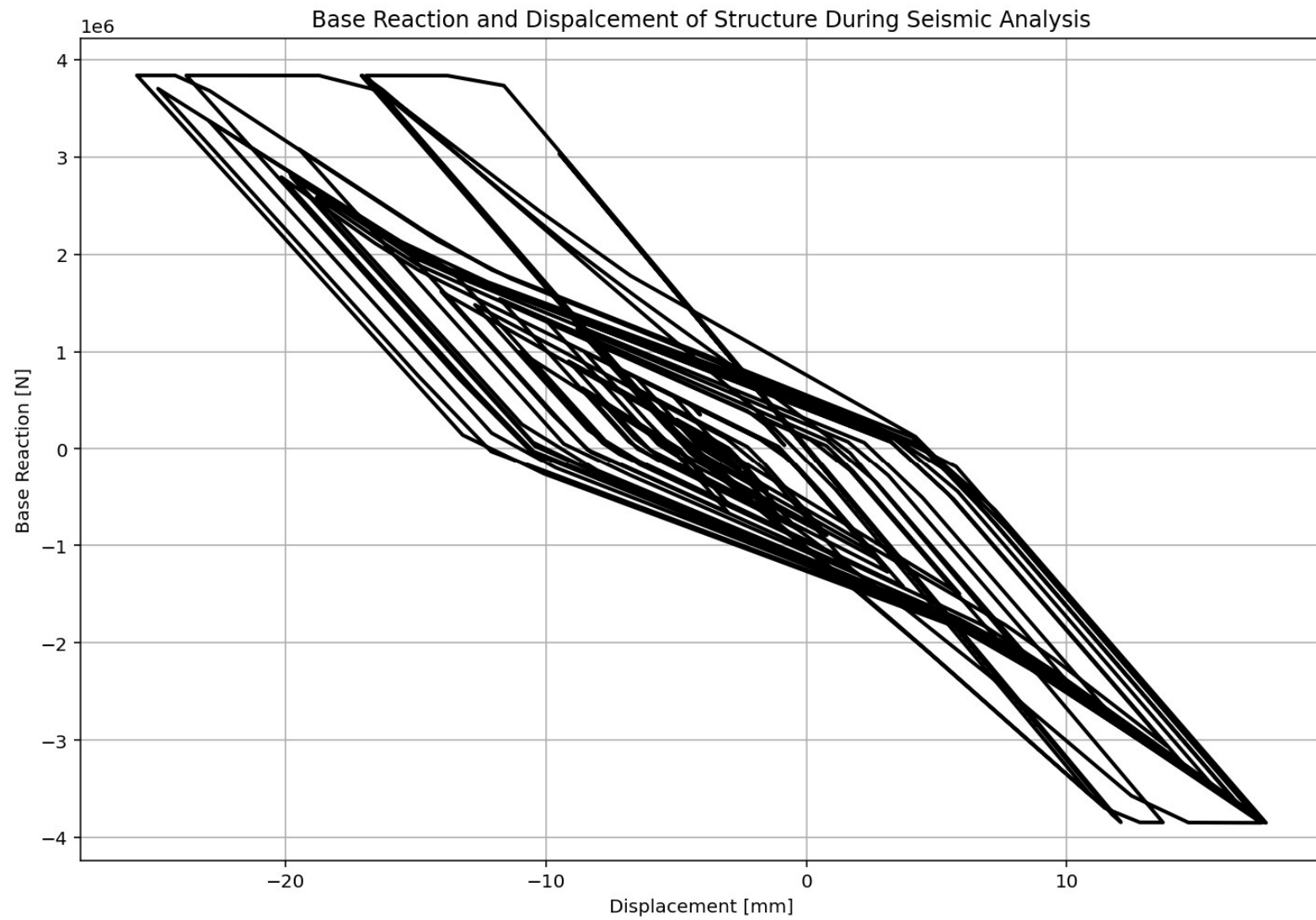
```
749 if ANAL_TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
750     ### DEFINE PARAMETERS FOR SEISMIC ANALYSIS
751     duration = 20.0           # [s] Analysis duration
752     dt = 0.01                # [s] Time step
753     GMfact = 9810            # [mm/s^2] standard acceleration of gravity or st
754     SSF_X = 10.50            # Seismic Acceleration Scale Factor in X Direct
755     SSF_Y = 10.50            # Seismic Acceleration Scale Factor in Y Direct
756     iv0_X = 0.0005           # [mm/s] Initial velocity applied to the node
757     iv0_Y = 0.0005           # [mm/s] Initial velocity applied to the node
758     st_iv0 = 0.0             # [s] Initial velocity applied starting time
759     SEI = 'Y'                # Seismic Direction
760     DR = 0.03                # Damping ratio
761
762     # Define time series for input motion (Acceleration time history)
763     if SEI == 'X':
764         SEISMIC_TAG_01 = 100
765         ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceler
766         # Define load patterns
767         # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fac
768         ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01,
769     if SEI == 'Y':
770         SEISMIC_TAG_02 = 200
771         ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceler
772         ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02,
773     if SEI == 'XY':
774         SEISMIC_TAG_01 = 100
775         ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath', 'Ground_Acceler
776         # Define load patterns
777         # pattern UniformExcitation $patternTag $dof -accel $tsTag <-vel0 $vel0> <-fac
778         ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', SEISMIC_TAG_01,
779         SEISMIC_TAG_02 = 200
780         ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath', 'Ground_Acceler
781         ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', SEISMIC_TAG_02,
782     print('Seismic Defined Done.')
```

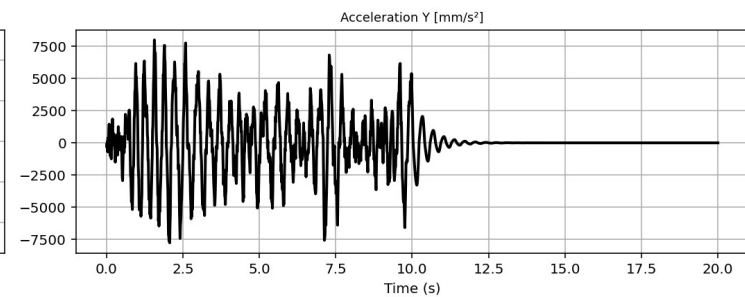
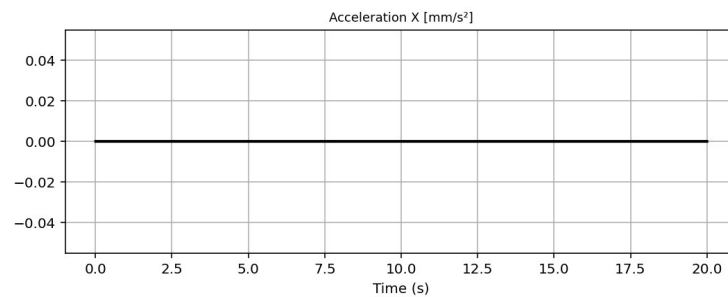
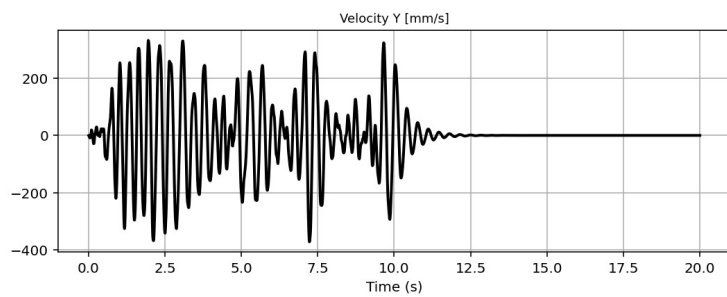
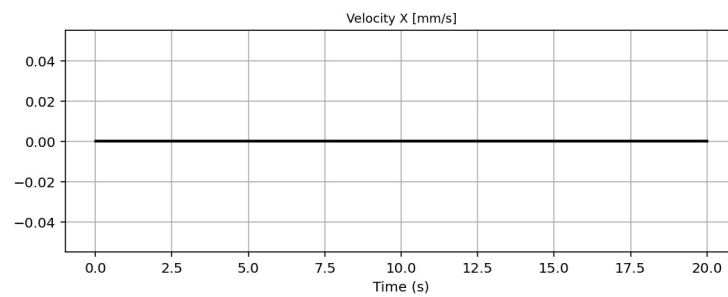
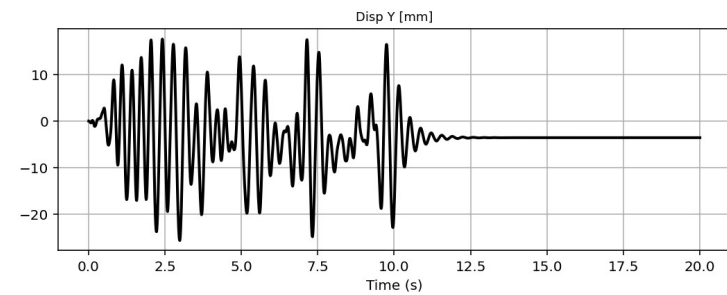
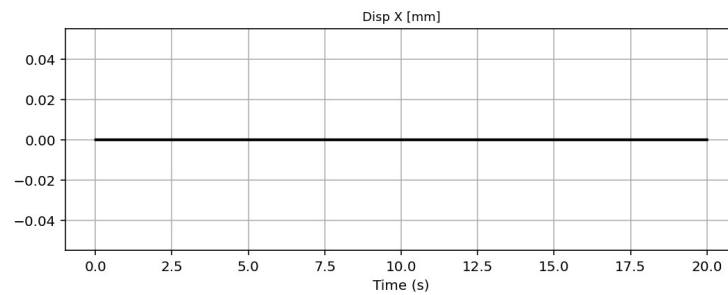
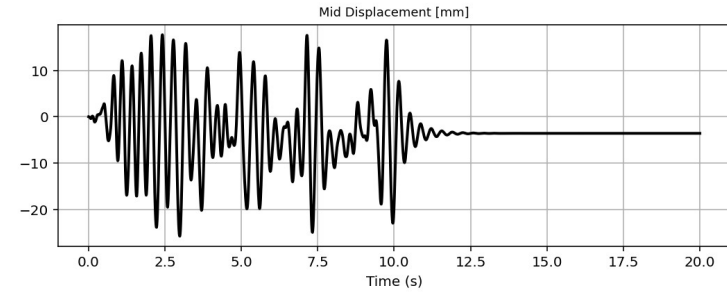
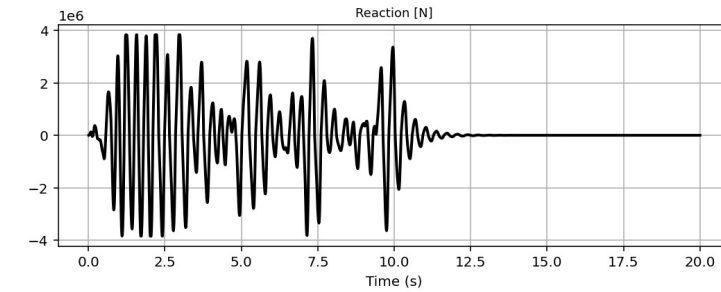
Base Reaction and Displacement of Structure During Seismic Analysis

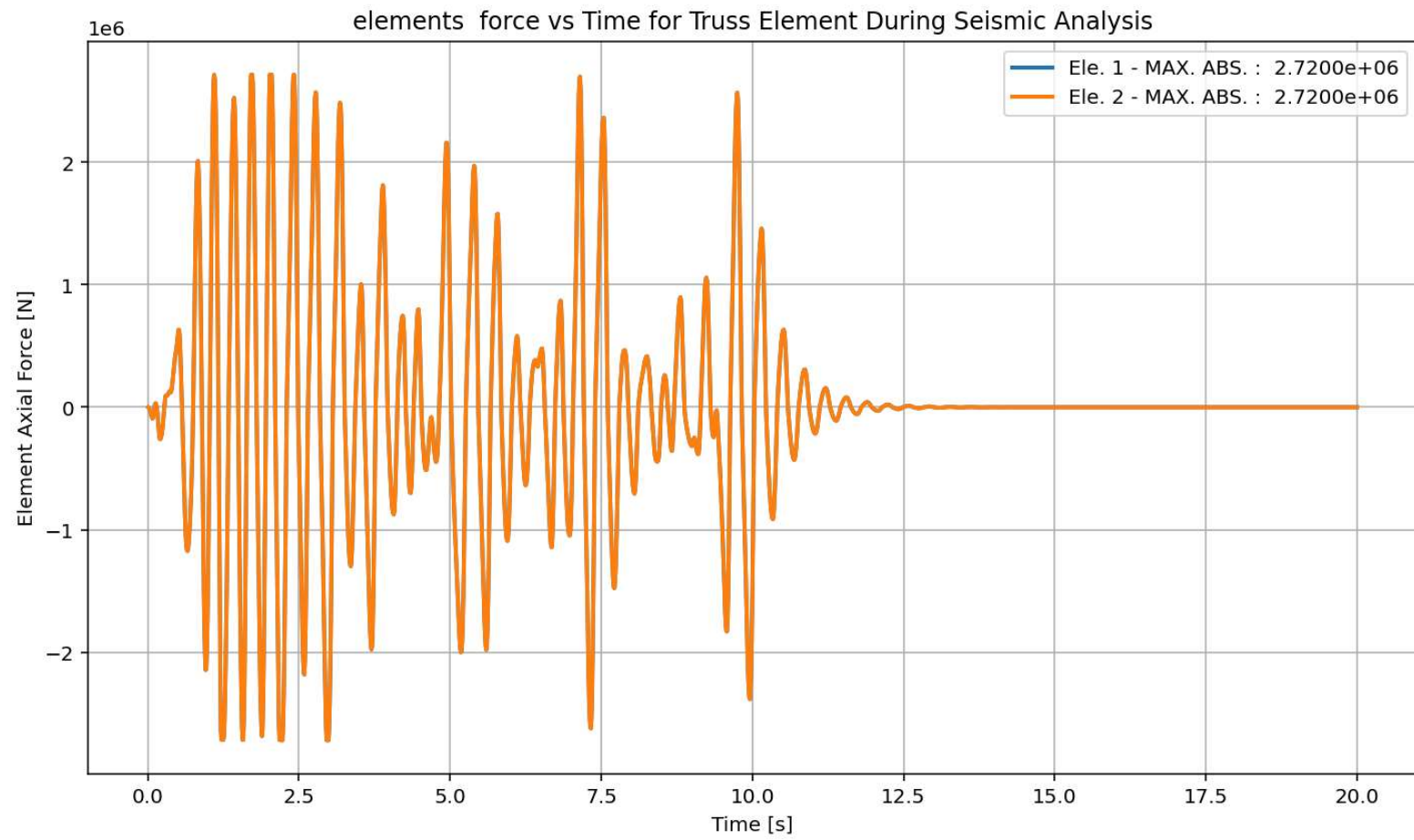


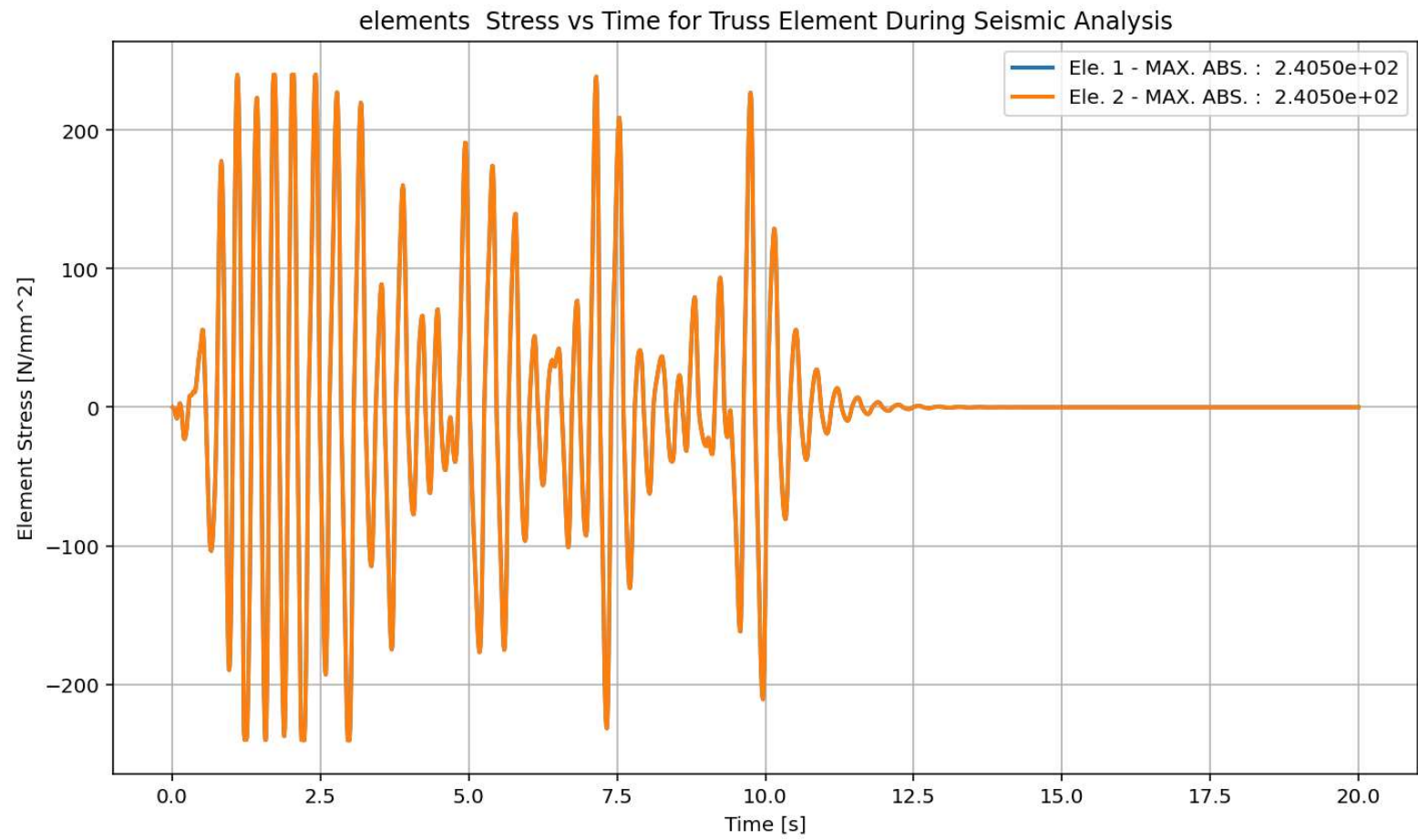
IPython Console Files Help Variable Explorer Debugger Plots History

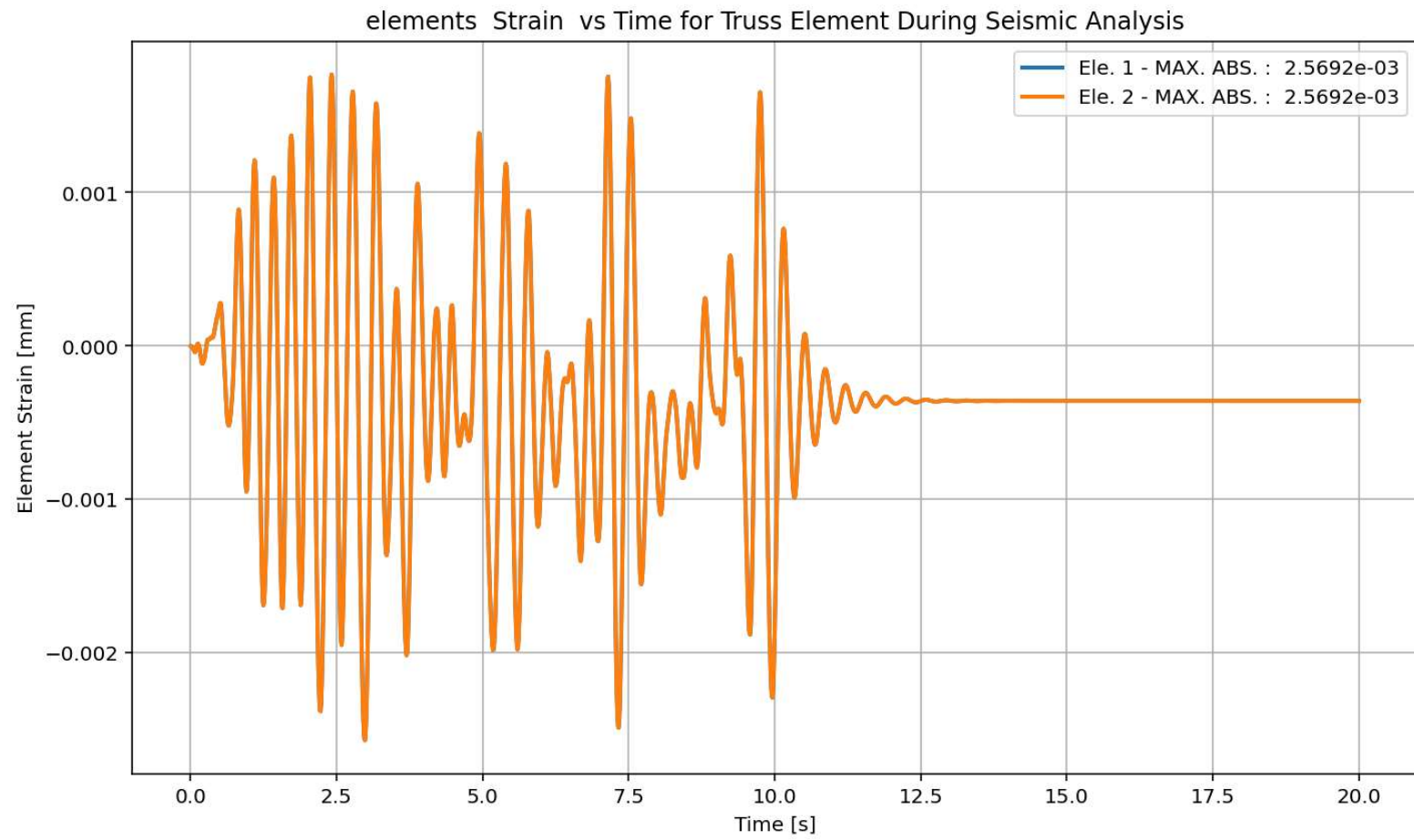
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 755, Col 44 UTF-8 CRLF RW Mem 52%

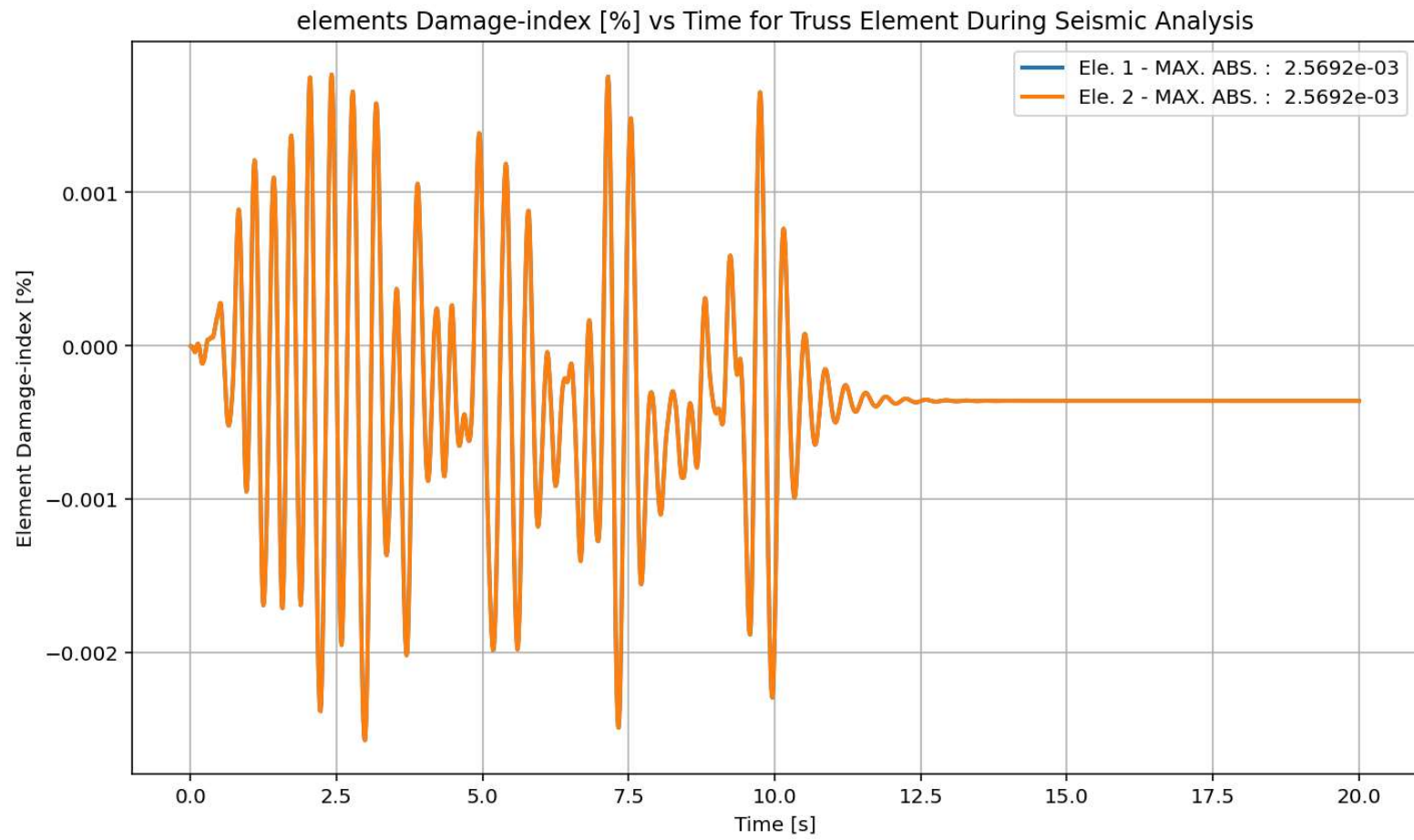


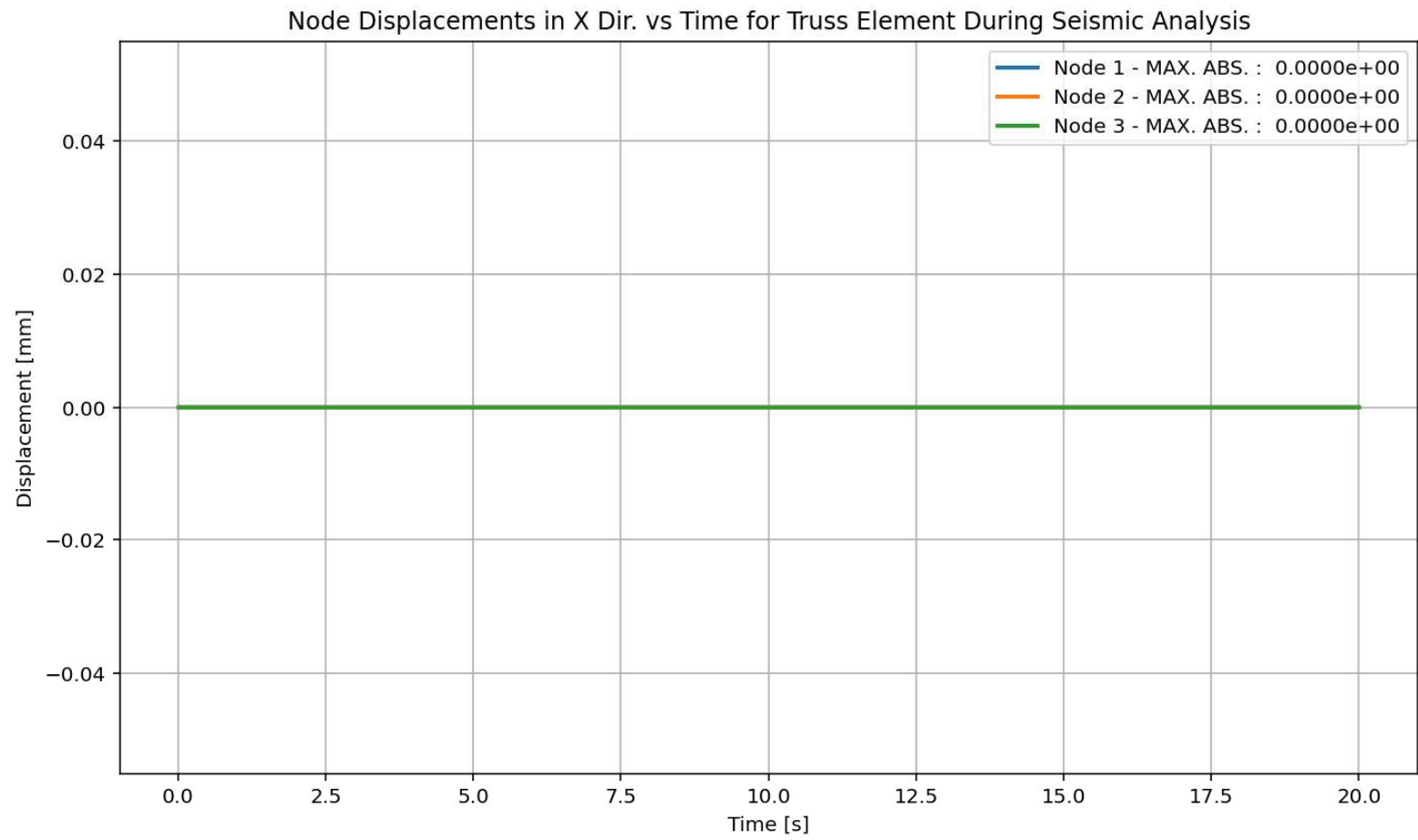


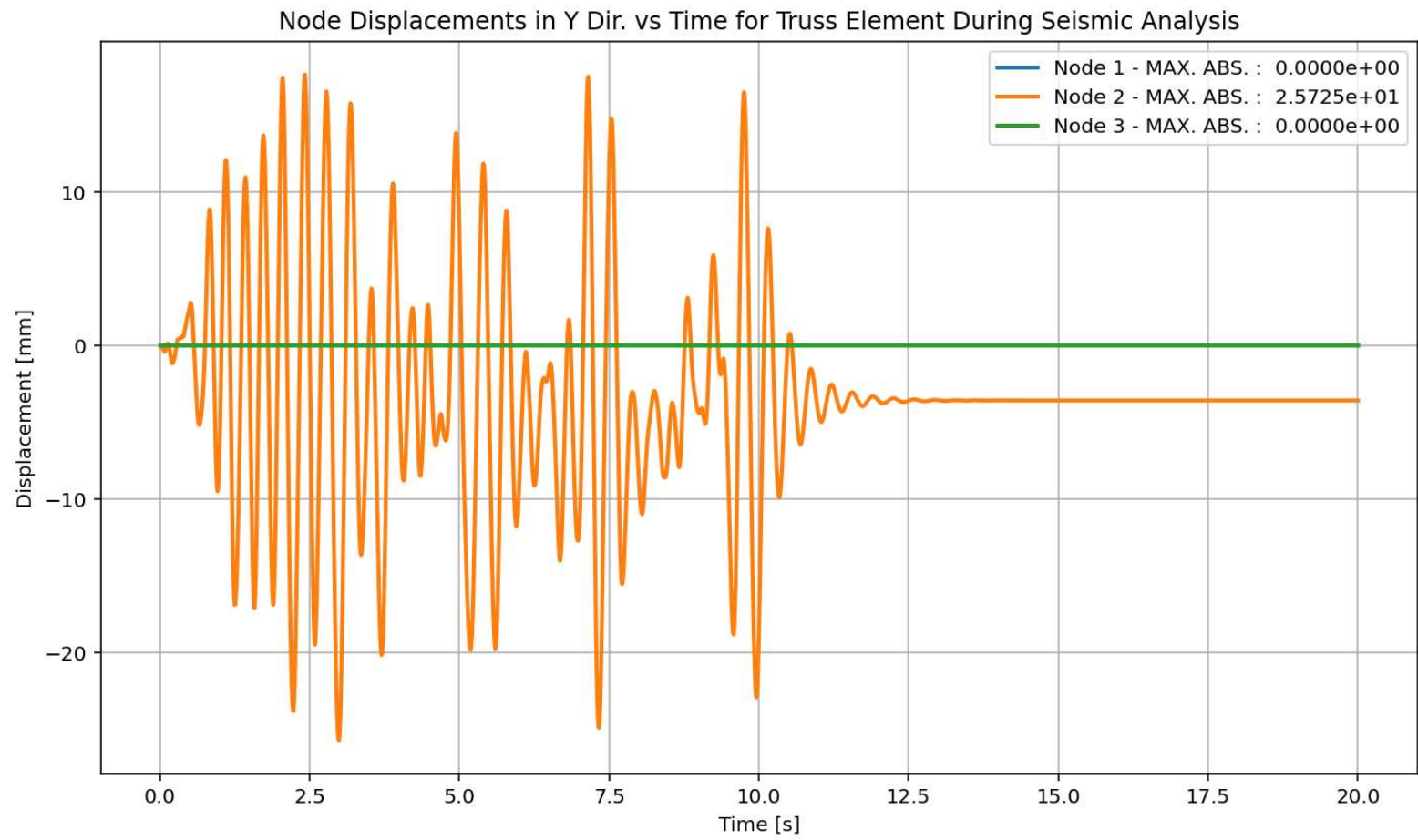




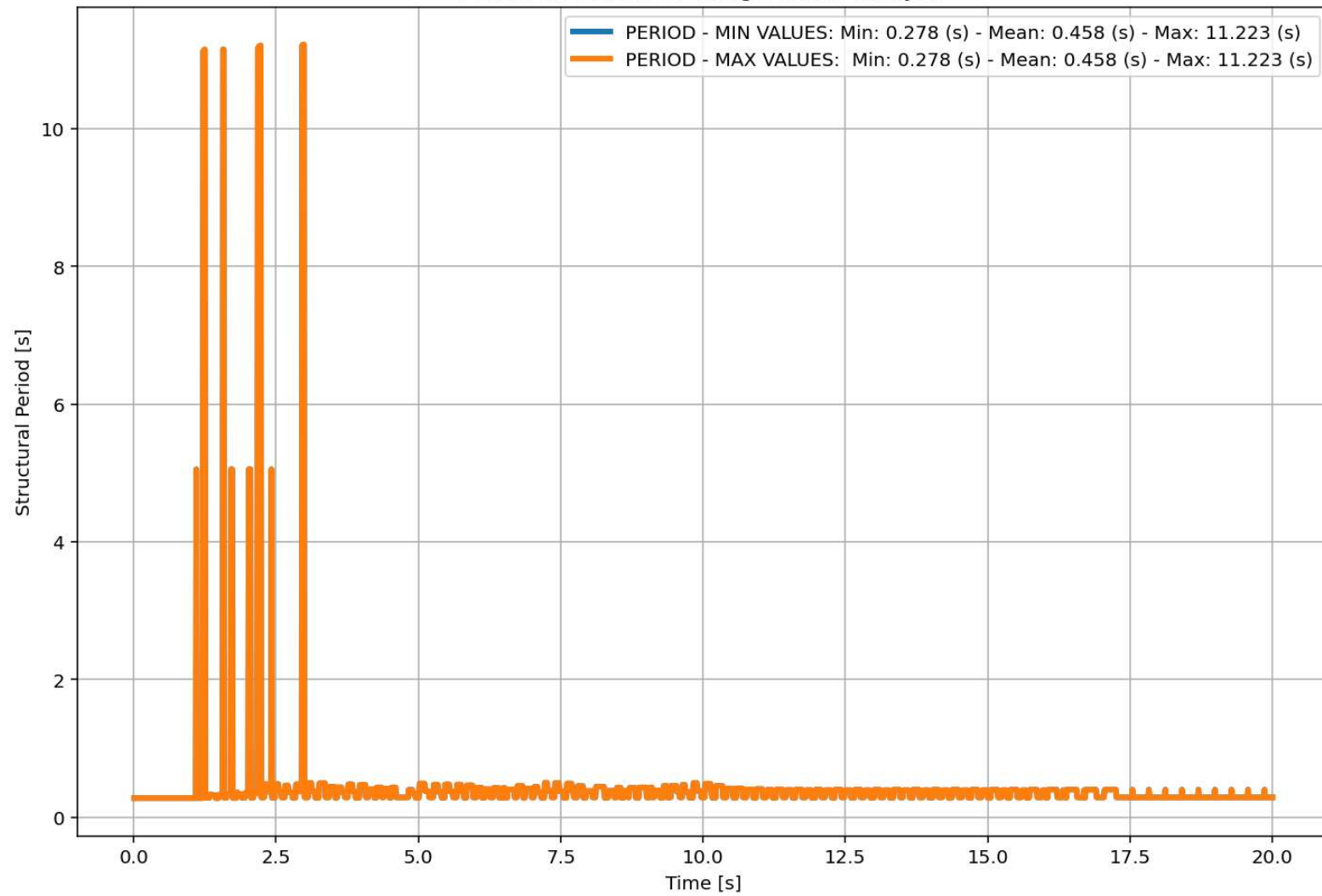


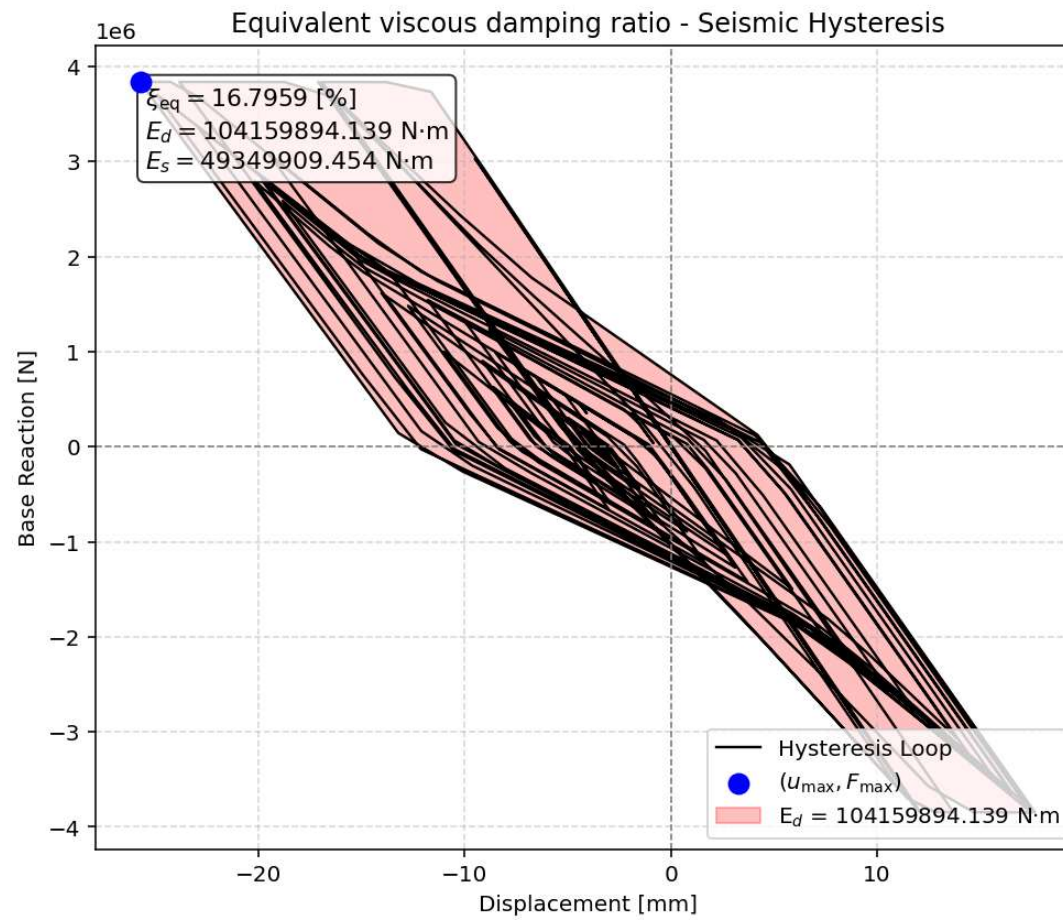


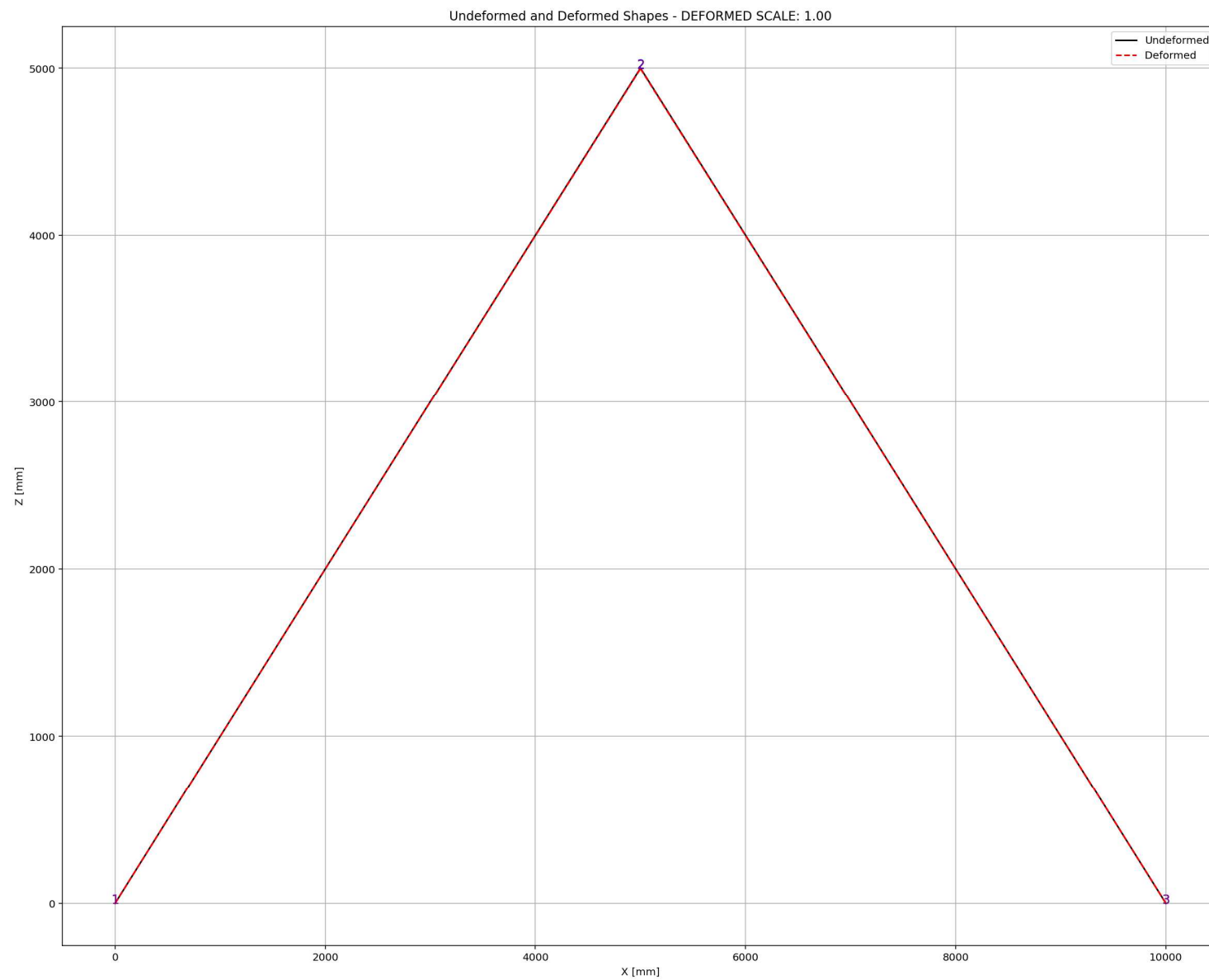


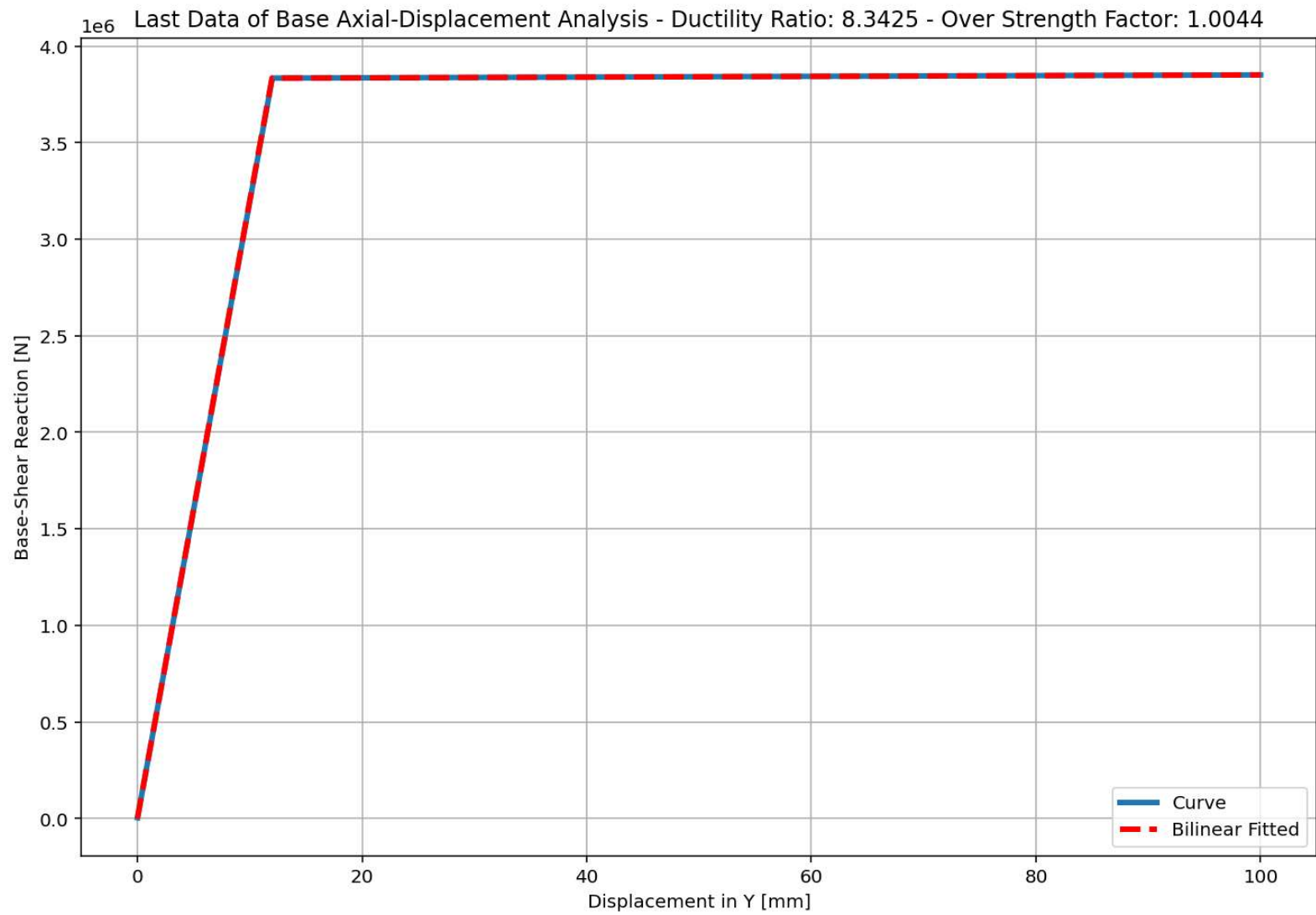


Period of Structure During Seismic Analysis









+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[1.19868464e+01 3.83386222e+06]

[1.00000000e+02 3.85055817e+06]]

+=====+

+-----+

Structure Elastic Stiffness : 319839.10

Structure Plastic Stiffness : 38505.58

Structure Tangent Stiffness : 189.70

Structure Ductility Ratio : 8.34

Structure Over Strength Factor: 1.00

+-----+

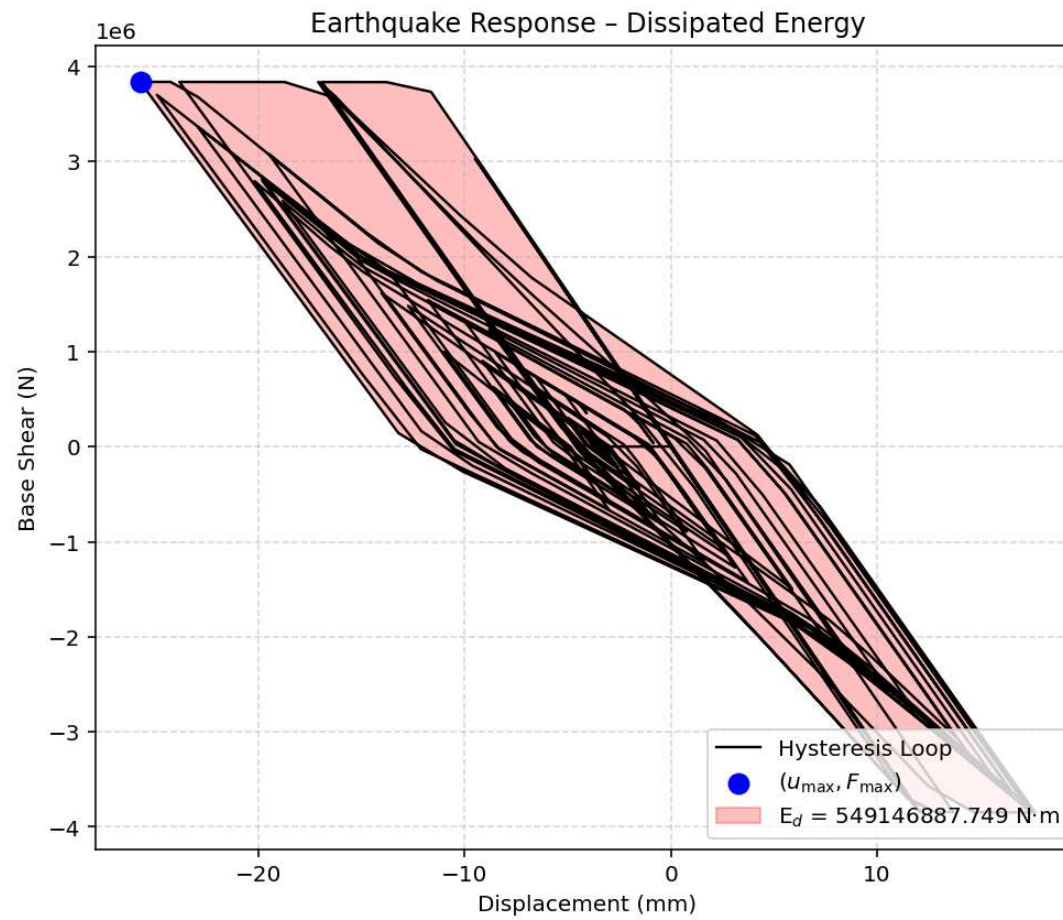
Over Strength Coefficient (Ω_0): 1.0044

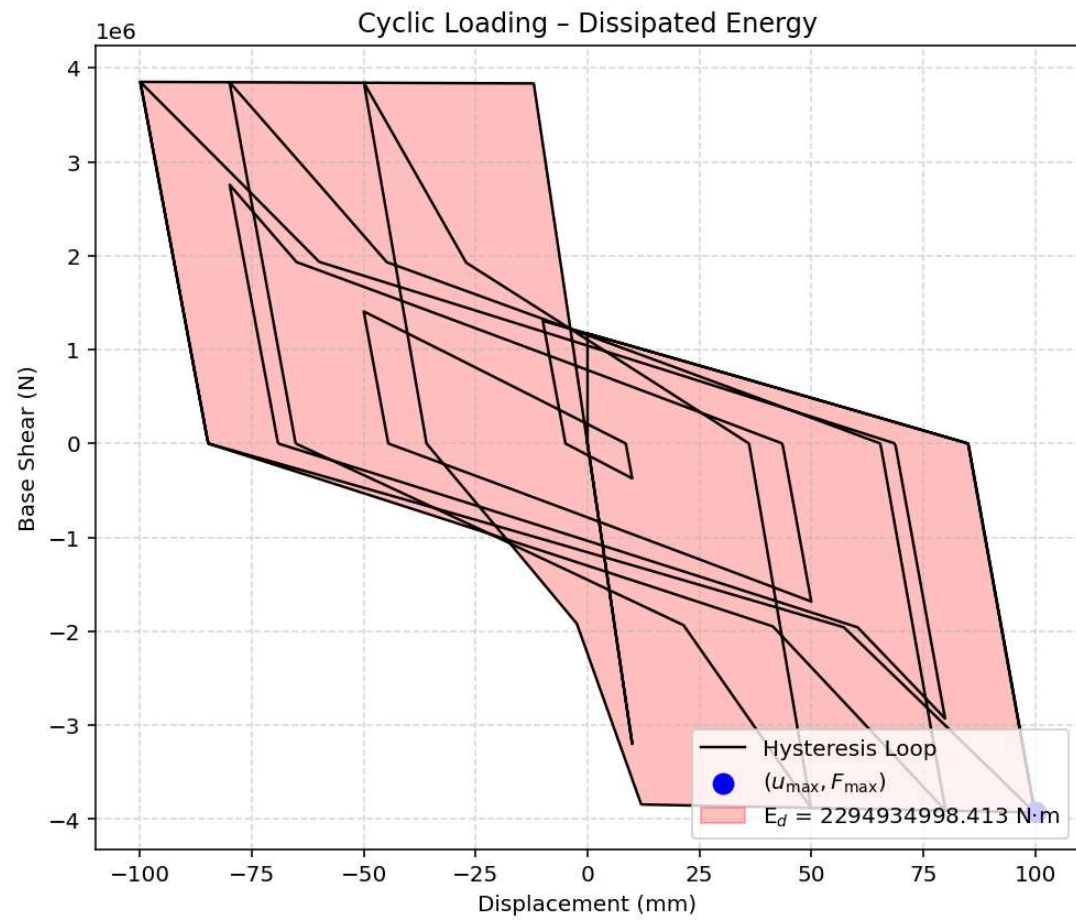
Displacement Ductility Ratio (μ): 8.3425

Ductility Coefficient (R_μ): 8.3425

Structural Behavior Coefficient (R): 8.3788

Structural Ductility Damage Index in Y Direction: 15.6090 (%)





Dissipated Energy from Earthquake= 549146887.75 N·mm
Dissipated Energy from Cyclic Displacement= 2294934998.41 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 23.929 [%]
ZONE 3: MODERATE-LOW DAMAGE

